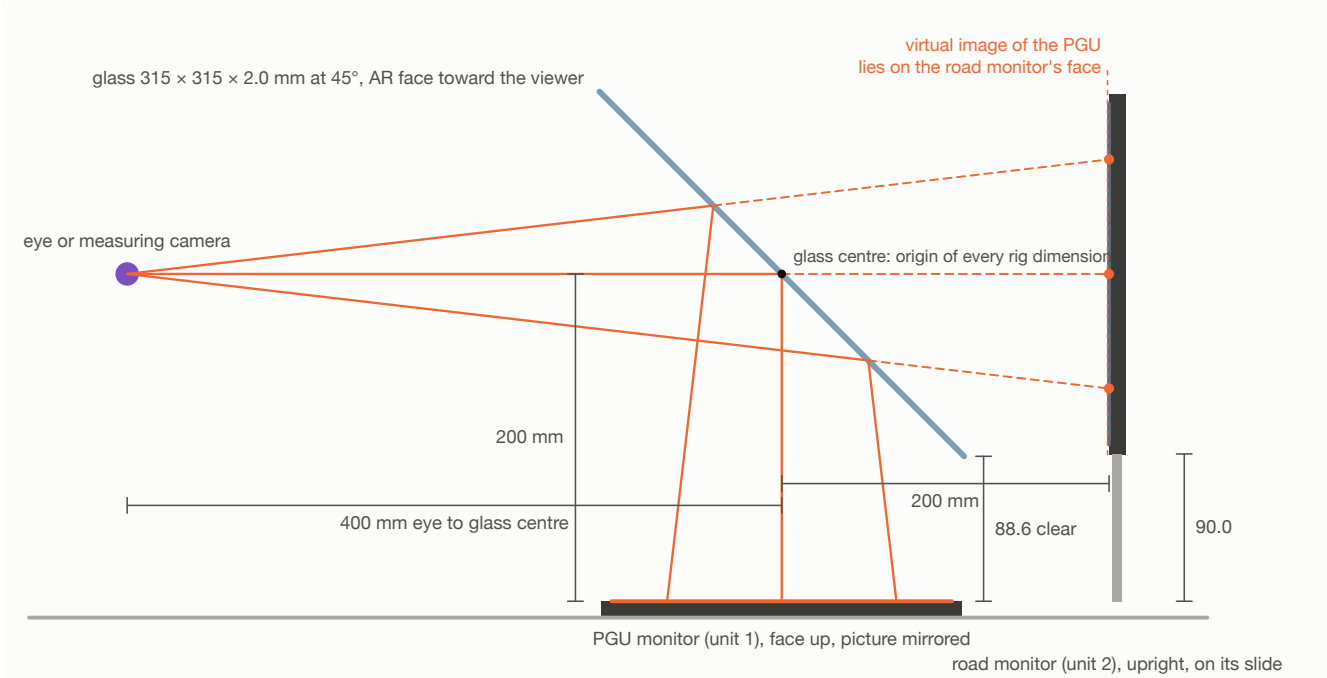
WINDSHIELD AR HUD · HUD-ONLY TFT LCD EDITION

Tabletop Design

Two 16-inch monitors and a sheet of teleprompter glass: one monitor shows a road, the HUD software detects the road in that picture, and the other monitor's reflection lays the registered overlay onto it. The design of the rig, the detection, the display chain, their calibration, and the tests that prove each part before anything goes into a car.



Eye to image 600 mm. Orange: light from three PGU points to the eye, with its virtual continuation (dashed) to the image on the road monitor's face.

Edition of 2026-10-27. Extends build stage B6 (Stage 0) of `docs/ARHUD-BUILD-GUIDE.pdf` and follows the plan in `docs/ARHUD-PLAN-TFT.pdf` and its specification `docs/ARHUD-PLAN-TFT-SPEC.md`. Shared numbers are imported from the plan's `calc.py`; bench numbers are derived in `docs/arhud-tabletop-src/tcalc.py`. Source of this document: `docs/arhud-tabletop-src`.

BEFORE ANYTHING ELSE

What this document is

The request was to build the tabletop display with the two YODOIT monitors and to design, in detail, the detection of the road and the display on the monitor that is reflected. This document reads that request one way, states it, and designs to it: **monitor unit 1 lies face up under the glass as the picture generator (the PGU), exactly as build stage B6 places it; monitor unit 2 stands upright behind the glass, where B6 had a black backdrop, and shows the road.** The HUD software runs its perception on the same road frames that unit 2 is showing, so the overlay the viewer sees in the glass is drawn from a road the software actually detected, registered to that picture. The result is the plan's whole software chain, from camera frame to lit pixel, running on a desk and measurable to a fraction of a degree, weeks before a car, a stereo camera or a custom mirror exists.

It is a design, not a build log: nothing here has been measured on the rig yet, and the document says at each number whether it is derived, assumed or still to be read off the hardware. It does not replace B6; it extends it, and where it departs from B6 it says why.

How to read it

Part I is the rig: its geometry, the light through the glass, what the viewer sees and how bright. **Part II** is the detection: where the road pictures come from, how lanes, vehicles and pedestrians are found in them and lifted to the car frame, and how that is scored. **Part III** is the display: how the renderer puts a car-frame point onto the PGU so that its reflection lands on the right pixel of the road picture, the timing of the chain, the calibrations, and the camera-in-the-loop variant. **Part IV** is the build: the software layout, the bring-up stages T-A to T-I with their gates, the risks, the bill of materials, and the appendices.

Conventions

MARK	MEANING
ASSUMED	A design value that a named measurement must replace. The text says which measurement and which stage makes it.
VERIFY	Not confirmed from a primary source, or not yet tested on this hardware.
DECISION	A choice the builder makes at that point; the options and their consequences are given.
☐	A check to tick before the stage's gate is run.
Frames	Rig frame: x along the viewing axis (the eye at negative x), y to the viewer's right, z up, origin at the glass's centre; millimetres. Car frame as in the plan: x forward, y right, z down, origin at the road camera; metres. Image frames: pixels, u right, v down.
Names	"PGU" or "unit 1": the monitor that is reflected. "Road monitor" or "unit 2": the one seen through the glass. "Window": the plan's 10.8° × 5.4° HUD field. "Bench stage" T-A to T-I: a chapter of Part IV; build stages B1 to B14 are the build guide's; optical Stages 0, 1 and 2 are the plan's.
Prices	US dollars, read on vendor pages on 2026-10-27 unless noted. Not quotes.

DESIGN BASIS: WHERE THIS BENCH DEPARTS FROM BUILD STAGE B6

ITEM	B6 (STAGE 0)	THIS DESIGN, AND WHY
Behind the glass	Black backdrop	The second monitor, showing the road. The overlay is then registered to something the software detected, and registration can be measured in pixels against truth. The backdrop remains the fallback if unit 2 is needed elsewhere (then unit 2 is the audience display, as the plan has it).
Image distance	About 0.6 m, uncontrolled	Exactly the road monitor's distance: glass centre 200 mm above the PGU and 200 mm in front of the road monitor, eye 400 mm from the glass, image at 600 mm. Coplanar planes make the overlay parallax-free and its calibration a single homography.
Data into hudd	Synthetic logs, then car recordings	Live perception on the road frames being shown (lanes, vehicles, pedestrians), with the recording's own speed and yaw rate as `ego`; synthetic logs remain the first test of hudd alone.
Crop on the monitor	Any placement	The window drawn at life size: 113.9 × 56.9 mm, 1,242 × 621 px, 115 px per degree, centred in the glass's reflected area.
Latency test	LED into the ZED, photodiode on the panel	Two photodiodes: one on a timing patch of the road monitor, one on the PGU's. No camera is needed; the chain is road photon to HUD photon.
Host	Jetson, or a mini PC	The MacBook Pro M4, which already drives one of the monitors and can drive both; the Jetson has been unreachable since 2026-10-07 and has one HDMI output. The Jetson takes over the PGU later, with a second host for the road monitor and a shared clock.
Head tracking	Not on the bench	A parallax variant: the road monitor slides back 31.6 mm to reproduce the car's parallax between its 7.5 m image and a road point at 20 m, and a webcam tracker is tested against a camera on a rail.

CONTENTS

PART I · THE RIG

1 What the tabletop proves, and what it cannot 6

2 Geometry and parts 8

3 Light through the glass 12

PART II · DETECTION

4 Road sources 18

5 Detection design 21

PART III · DISPLAY

6 The renderer on the bench 26

7 Clocks and latency 29

8 Calibration 31

9 Camera in the loop 34

PART IV · BUILD

10 Software layout 37

11 Bring-up stages and gates 39

12 Risks, gotchas and the bill of materials 42

A Formulas 44

B The calibration file 45

C Sources and provenance 46

PART I

The rig

Two monitors, one sheet of glass, one viewing position. What the viewer sees, how large, how bright, and why the two pictures lie on one plane.

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1 What the tabletop proves · 2 Geometry and parts · 3 Light through the glass

1 What the tabletop proves, and what it cannot

A head-up display has two halves that can be wrong independently: the perception that decides where the lane and the cars are, and the display chain that puts a mark on the glass where the driver's eye will see it over that lane or car. The car tests both at once and blames neither. The tabletop separates them. The road is a picture whose every pixel is known, so perception can be scored against truth; the overlay is a reflection onto that same picture, so the display chain can be measured in pixels. Everything between the camera frame and the lit pixel is the plan's own software, unchanged.

1.1 The two monitors' roles

Unit 2, the road monitor, stands upright behind the glass and shows the road: a recorded drive (a comma2k19 segment, an openpilot route), a MetaDrive render with full truth, a phone's dashcam clip, or later a live camera. The picture is shown at life size, meaning the monitor subtends at the eye the same angle that the shown part of the frame subtended at the camera, so that the window drawn over it is the plan's 10.8° × 5.4° window at its real angular size. The viewer looks through the glass at this monitor.

Unit 1, the PGU, lies face up under the glass, as in B6. The glass reflects it toward the viewer, and because the glass is as far above unit 1 as it is in front of unit 2, the reflection appears exactly on unit 2's face. The renderer draws the HUD window, mirrored, into a 1,242 × 621 px region of unit 1 and leaves the rest black; the viewer sees the overlay floating on the road picture.

Perception runs on the frames unit 2 is showing. In the direct path they go from the player straight to the lane and object networks, with each frame's "capture time" defined as the moment it is on the road monitor; in the camera-in-the-loop path (chapter 9) a camera above the glass looks at unit 2 and the networks see what it sees. Either way the detected lanes and objects are lifted to the car frame on the recording's camera model, published as the plan's `lanes` and `objects` topics, age-compensated with the recording's speed and yaw rate, and projected by `hudd` onto the PGU through a calibrated mapping, so that the drawn line falls on the painted line as the viewer sees it.

1.2 What it proves

- **The perception chain on real road pictures**, scored against truth (MetaDrive) or a reference (openpilot's own lane model on its routes), with the plan's §5.7 criteria, before the ZED exists. Monocular only: the plane is static, as in the plan's fallback mode.
- **The renderer's projection, drawing rules and warp** to a tenth of a degree: a drawn lane line against the painted line it came from, measured from the eye point. With latency removed and `hudd` fed the picture's truth instead of perception (truth-fed sync mode, chapter 6) every error is calibration or drawing; with perception in the loop the same measurement is the whole chain's.
- **Latency and age compensation**, live: perception runs on a frame only once that frame is on the road monitor, so the chain from road photon to HUD photon is the car's chain with the camera replaced by a monitor. The prediction's sign is confirmed on a curve (B6.5) with real speed and yaw rate from the recording, and the chevron's residual with and without compensation is measured against the picture's truth.

- **Contrast as the driver will see it.** The road monitor's brightness and the renderer's gain reproduce the plan's contrast ratios (1.15 minimum, 1.19 and 1.39 for the design backlight) at the eye, so the drawing rules (ribbon fill share, line widths, night level) can be judged under the right ratios, if not the right absolute levels.
- **Head tracking and parallax**, in the variant where the road monitor slides back: the car's parallax between its 7.5 m image and a 20 m road point is reproduced by a 31.6 mm offset, and the tracker's correction is measured on a rail.
- **The software itself.** Every process, topic, config file and tool written for the bench is the car's; only `roadsrc` (the player) is bench-only, and it is replaced by `roadd` (the ZED) in the car.

1.3 What it cannot prove

- Nothing about the relay optics: no magnification, no windshield, no mirror figure, no eye box, no image distance beyond 0.60 m. Stage 1 and Stage 2 remain the only tests of those.
- Daylight brightness. The brightest background the rig can show is the road monitor's white through the glass, about 840 cd/m², against a sky of 10,000. Ratios are reproduced; adaptation and absolute levels are not.
- Stereo depth and the per-frame plane fit. A camera looking at the monitor sees a flat plane at half a metre. Range on the bench comes from the ground plane at the box base, as in the plan's fallback; the stereo path is tested only in the car.
- The eye-side range error. The plan's registration model (§5.4) puts the camera 45 cm from the eye, so a range error moves the mark; on the bench the eye is at the camera by construction (both are the recording's camera), so that term is zero and must be measured in the car.
- Sun, heat, vibration, the controller, the backlight, the shutter: none of them is on the bench.

1.4 The fallback configuration

If unit 2 is wanted elsewhere, the rig reverts to B6 as written: black backdrop behind the glass, unit 2 as the audience display, perception from recordings without a picture to register to. Every other chapter still applies; only the registration measurements against the picture (T-E, T-H, T-I) are lost.

2 Geometry and parts

Every dimension of the rig follows from one rule: the glass's centre is the same distance from the PGU's face as from the road monitor's face. Then the reflection of the PGU lies on the road monitor, the two pictures are one plane, and the viewer may sit anywhere in front of the glass without the overlay sliding.

2.1 The monitors

Both are the owned YODOIT 16-inch panels: 3,840 × 2,400 px, which the plan takes as 277 ppi (91.7 µm per pixel, 352.1 × 220.1 mm active, a 16.3-inch diagonal). A true 16.0-inch diagonal would be 282.8 ppi and 344.9 × 215.6 mm. ASSUMED Measure the active width with a rule in T-A; every millimetre in this chapter scales with it, and the pixel pitch enters the calibration file. The vendor's peak white is 1,200 cd/m² and the contrast 1,200:1 or 2,000:1 depending on the listing VERIFY; both are metered in T-G. One of the two is attached to the MacBook Pro today and reports 3,840 × 2,400 at 60 Hz; the Jetson drove the other over HDMI on 2026-10-07, so HDMI input exists on at least that unit.

2.2 The glass

The Neewer NA17 teleprompter beamsplitter: 315 × 315 × 2.0 mm tempered glass, 70% transmitted and 30% reflected, with an anti-reflection coating on one face, about $53. It is the cheap one of the plan's pair and the one B6 names. At 45° its footprint along the viewing axis and its height are both 222.7 mm, which just covers the PGU's short side (220.1 mm); its width is 37.1 mm less than the monitors' 352.1 mm, so 18.6 mm of the PGU at each side is never reflected and 3,435 px of its 3,840 are usable. The HUD window needs 113.9 mm; the margin is large. If the larger glass the plan lists for Stage 1 is bought instead (a 30R/70T sheet of about 16 × 21 inches), the whole PGU is reflected; nothing else changes.

Which face carries the beamsplitter coating and which the anti-reflection coating is not stated on the vendor page. DECISION in T-A, by test: photograph a thin bright line on the PGU through the glass in both orientations and keep the one with the fainter second image (section 3.6).

2.3 Dimensions

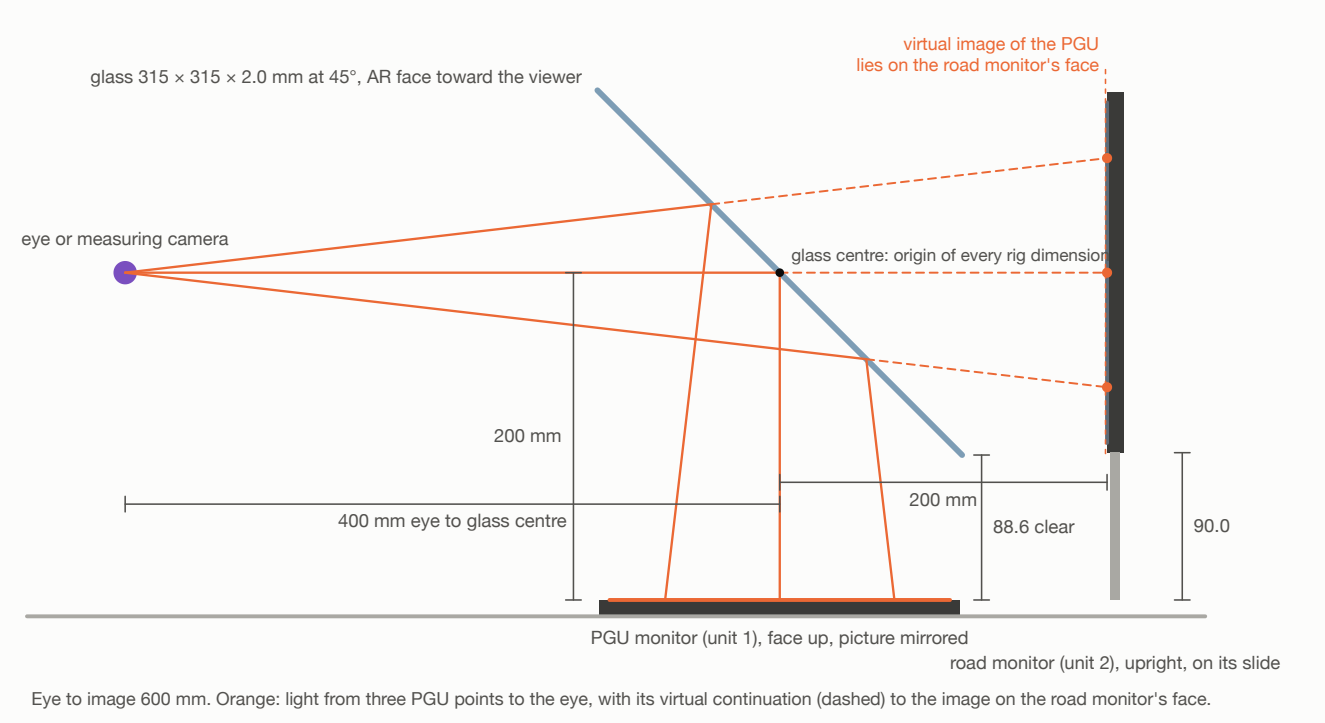


Figure 2.1. Side view to scale. Rig frame: x along the viewing axis, z up, origin at the glass's centre. The road monitor's lower edge is 90.0 mm above the PGU's face level, so it stands on a 90.0 mm block or its own stand.

DIMENSION	MM	WHY
Glass centre above the PGU's face	200	Any value above the glass's own half-height plus clearance; 200 keeps B6's 0.6 m image and leaves 88.6 mm between the glass's low edge and the PGU.
Glass centre to the road monitor's face	200	Equal to the first: the coplanar condition. Tolerance 2.8 mm (section 3.1).
Glass tilt from horizontal	45°	Reflects the vertical from the PGU into the horizontal toward the eye; ± 0.5° with the angle gauge. The top edge leans toward the viewer.
Eye to the glass centre	400	Sets the image distance to 600 mm and the life-size field to 32.7° × 20.8°; other distances in Table 3.1.
Eye height	0	Level with the glass's centre and the road monitor's centre.
Glass low edge above the PGU face	88.6	Clearance for the monitor's bezel and cable.
Glass top edge above the PGU face	311.4	The loop camera (chapter 9) mounts just above and behind it.
Road monitor's lower edge above the PGU face level	90.0	Centres the road picture on the glass's centre.
Line of sight through the glass to the road monitor's top and bottom edges crosses the glass at	62.0 / -89.8	Within the glass's ± 111.4 mm, so the whole road monitor is seen through the glass from the eye point (yes).

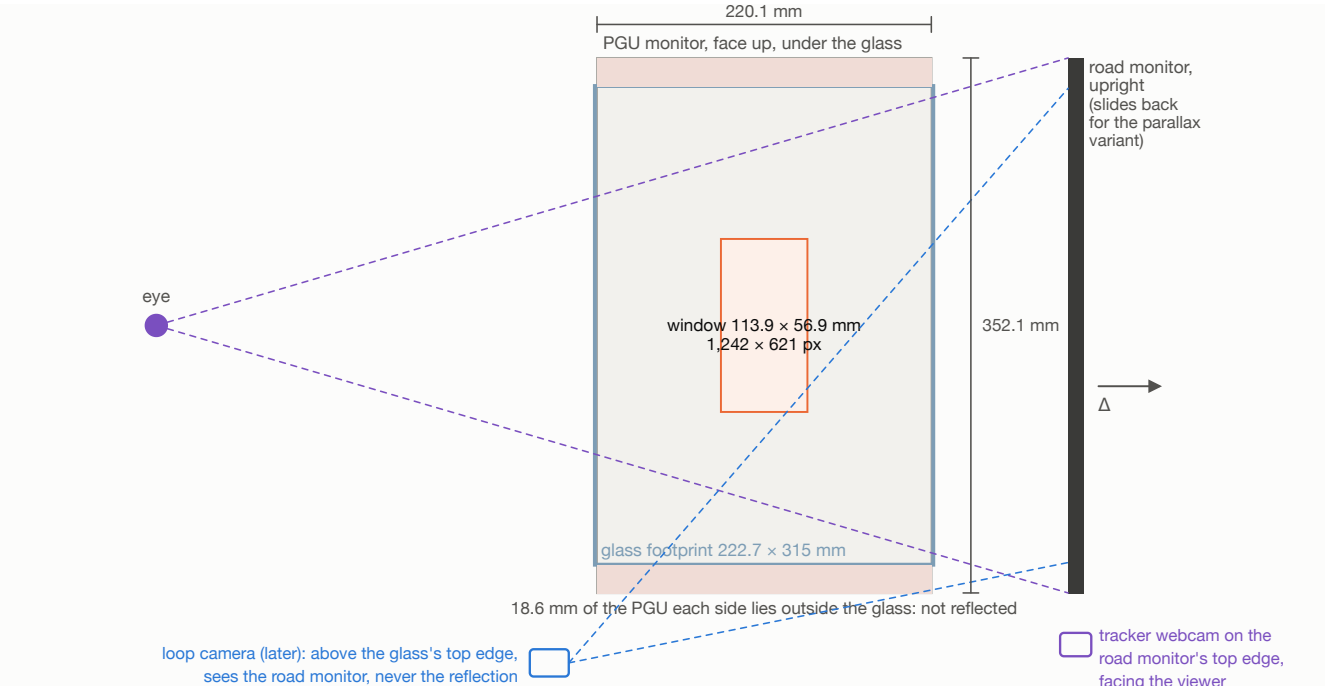


Figure 2.2. Plan view to scale. The glass's footprint, the PGU under it with the window at its centre, the strips at each side that the glass does not reach, the road monitor on its slide, and the positions of the two cameras that later stages add.

2.4 Enclosure, stands and the slide

B6's enclosure serves with two changes. The backdrop becomes the road monitor's stand. And the glass cannot ride in slots cut into B6's cheeks: those stand 352.1 + 10 mm apart so the PGU fits between them, and the NA17 is only 315 mm wide. So a base of 6 mm plywood or black foam board about 750 × 450 mm; two cheeks 352.1 + 10 mm apart with a well between them where the PGU lies face up on felt, its cable out the back; and on each cheek's inside face a 23 mm spacer block carrying a grooved batten at 45°, the two grooves 315 + 2 mm apart, so the glass seats with its centre 200 mm above the PGU's face. The battens start 88.6 mm above the PGU's face, so the spacer blocks clear the monitor. (The larger glass the plan lists for Stage 1 would reach B6's cheek slots directly.) Every inside face matte black or flocked. Behind the glass, a 300 mm drawer slide or a camera macro rail carries the road monitor on a block 90.0 mm high, with a millimetre scale along the slide and a stop at the coplanar position, so the parallax variant (section 3.8) is a slide to a number and back. A hood of black card over the top keeps room light off both monitors and off the glass.

At the eye point, a chin rest or a tripod with a quick-release plate at the glass centre's height, 400 mm from it. The measuring camera (a phone with manual exposure, or a mirrorless camera) mounts there for every calibration and test; the viewer sits there for the demo.

2.5 Hosts and cables

The MacBook Pro M4 drives both monitors from the start: it is attached to one now and the model supports two external displays VERIFY. Both run at native 3,840 × 2,400 and 60 Hz, as separate displays, with macOS scaling off for the windows that matter (chapter 10); auto-brightness, True Tone and Night Shift off for the externals. The monitors' own power: a USB-C PD charger each, because a laptop port may not sustain full

brightness VERIFY. Every process of the bench runs on the Mac on one clock. When the Jetson returns it takes the PGU over HDMI and runs perception; the Mac keeps the road monitor and `roadsrc`, and the two hosts share a clock over Ethernet (chapter 7).

2.6 Assembly

1. Measure both monitors' active width and height with a rule; compute the pixel pitch; write it into `config/calibration.yaml`.
   - ☐ Pitch recorded for both units
2. Build the base, cheeks and hood as in B6.2, the cheeks 352.1 + 10 mm apart. Glue a 23 mm spacer block and a grooved 45° batten to each cheek's inside face, grooves 315 + 2 mm apart, placed so that the seated glass's centre is 200 mm above the PGU's face and its lowest edge 88.6 mm above it. Flock or paint every inside face.
3. Fit the slide behind the glass, centred, its scale zeroed where the road monitor's face will be 200 mm from the glass centre. Fit the 90.0 mm block to the slide's carriage.
4. Lay the PGU face up in the well, its top edge away from the viewer, level; slide the glass into the battens, AR face toward the viewer for now; stand the road monitor on its block at the zero mark.
   - ☐ Glass 45° ± 0.5°; PGU level; road monitor vertical
5. Measure the glass centre to the PGU's face and to the road monitor's face with a rule; adjust the block or the stop until both read 200 mm within 1 mm.
6. Connect both monitors, set native resolution and 60 Hz, disable auto-brightness; show the test card on the PGU and a grid on the road monitor.
   - ☐ Test card reads correctly in the glass (hudd's flip set)
7. Run the parallax check of section 3.1 from the eye point; shim until it passes.
   - ☐ Separation ≤ 5 px over a 100 mm shift

3 Light through the glass

A flat beamsplitter adds nothing to the picture but a mirror flip and a loss. What matters is where the reflection lands, how large the road appears, how bright each picture is at the eye, and what the glass's second face does to a thin line.

3.1 The coplanar condition and its check

A flat mirror puts the image of a point as far behind the mirror as the point is in front of it. With the glass's centre 200 mm above the PGU's face and 200 mm in front of the road monitor's face, every PGU pixel's image lies on the road monitor's face. Two consequences carry the whole design. First, there is no parallax: the overlay and the road are one plane, so the viewer may move and a mark drawn over a painted line stays over it. Second, the mapping from a road-monitor pixel to the PGU pixel whose reflection covers it is a fixed two-dimensional transformation, independent of where the eye is; for a perfect rig it is a vertical mirror at unit scale, and for the real one it is that plus a small homography that one photograph fits (chapter 8).

The condition is checked by parallax, which is far more sensitive than a rule. Draw a cross at the window's centre on the PGU and the same cross on the road monitor. From the eye point, photograph them; move the camera 100 mm sideways and photograph again. If the planes differ by δ, the two crosses separate by 100 × δ / 600 mm on the road monitor, 1.8 px per millimetre of δ. A tolerance of 5 px therefore holds the planes within 2.8 mm, closer than a rule reads on a monitor behind glass.

3.2 The flip

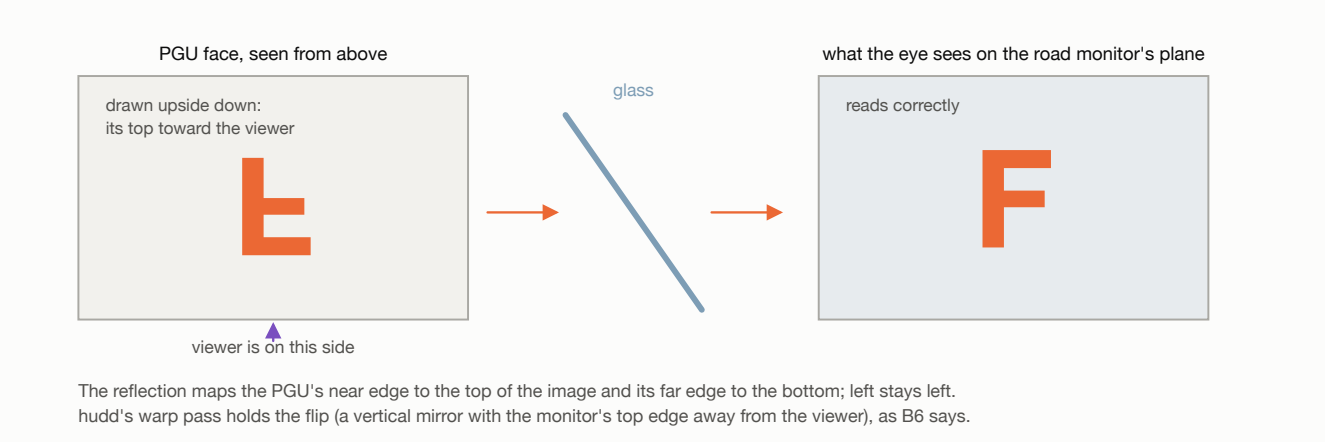


Figure 3.1. One reflection mirrors the picture. With the PGU's top edge away from the viewer the picture reads upside down; the renderer's warp mirrors it vertically.

Mirroring about the glass maps a PGU point at (x, y, −200) in the rig frame to (200, y, −x): the PGU's near edge becomes the top of the image, its far edge the bottom, and left stays left. Laid with its top edge away from the viewer (cable out the back), the PGU's picture reads upside down in the glass; `hudd` 's warp pass carries a vertical mirror. Rotating the monitor instead would trade it for a horizontal mirror; one of the two is unavoidable, as in any single reflection. The test card of B6.2 settles it.

3.3 Life size: the angular scale

The viewer at 600 mm sees the road monitor's 352.1 × 220.1 mm as 32.7° × 20.8°. If the road monitor shows the part of the camera frame that subtended the same 32.7° × 20.8° at the camera, every feature in the picture is at its real angular size, the plan's 10.8° × 5.4° window is 113.9 × 56.9 mm on the monitors (1,242 × 621 px), and the renderer's drawing rules are judged at their real angular scale. The window then has 115 px per degree against the plan's 87 on the 5.5-inch crop, so anything legible on the bench is legible in the car. One monitor pixel is 0.53′ from the eye; 0.1° is 11.4 px; the plan's 0.5° static criterion is 57 px and its 0.3° head-motion criterion 34 px.

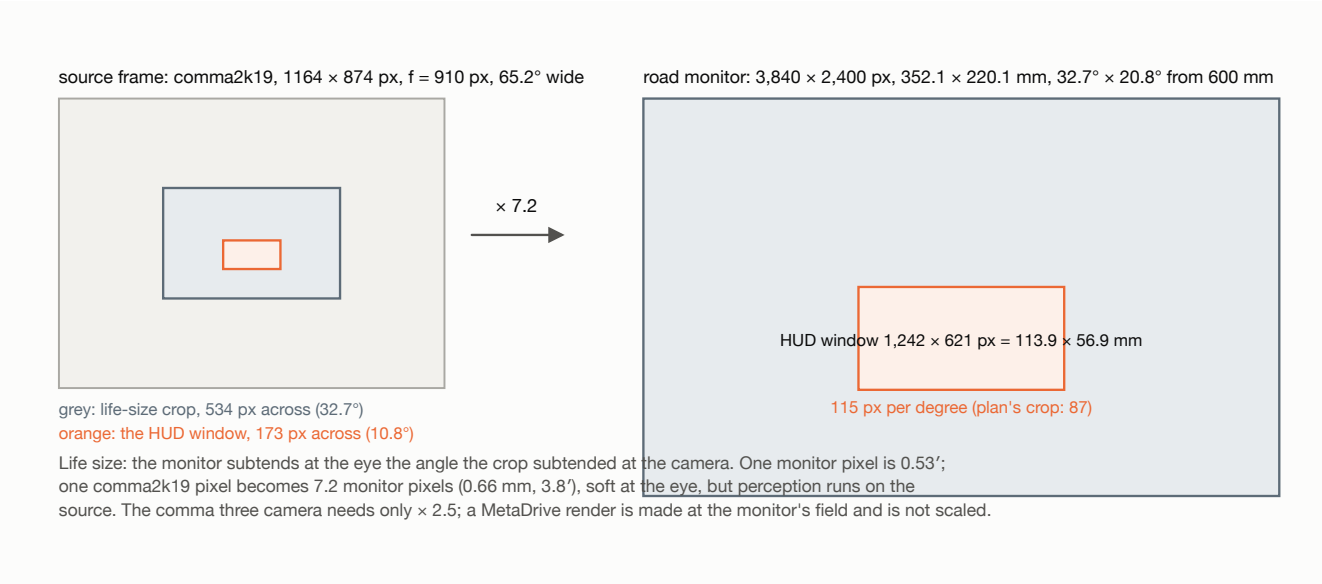


Figure 3.2. The life-size crop for a comma2k19 frame and what it becomes on the road monitor. Perception runs on the source frame, never on the enlarged picture.

The price is upscaling. A comma2k19 frame has only 534 px across the shown field, so each source pixel covers 7.2 monitor pixels (3.8′): the road picture is soft at the eye, though sharper sources exist (Table 4.1) and the lane detector sees the original pixels. The distance is a free parameter; Table 3.1 gives the others. **DECISION** in T-C: 600 mm is the design, chosen so that the plan's window keeps more than 87 px per degree while leaving 100.6 mm of glass beyond the window at each side and 82.9 mm front and back.

EYE TO IMAGE (MM)	EYE TO GLASS (MM)	ROAD FIELD SHOWN, LIFE SIZE	WINDOW ON THE MONITORS (MM)	WINDOW (PX)	PX PER DEGREE
450	250	42.7° × 27.5°	85 × 43	932 × 466	86
600 (design)	400	32.7° × 20.8°	114 × 57	1,242 × 621	115
750	550	26.4° × 16.7°	142 × 71	1,553 × 776	143
900	700	22.1° × 13.9°	171 × 85	1,863 × 932	172

Table 3.1. Eye-to-image distance against what the road monitor shows at life size and the size of the window on the monitors. The glass centre stays 200 mm from both monitors; only the eye moves.

3.4 Luminance and contrast at the eye

On the vendor's figures, the PGU's white reaches the eye as 1,200 × 0.30 = 360 cd/m², and the road monitor's white through the glass as 1,200 × 0.70 = 840 cd/m². A noon clip's sky is that white; its sunlit asphalt sits near 20% of it, 168 cd/m² ASSUMED. The HUD white over the clip's sky therefore gives a contrast ratio of at most 1.43, and over its asphalt 3.14. Both exceed the plan's conditions, so the renderer reproduces a chosen condition by drawing at a **gain**, the share of the PGU's white that makes the ratio at the eye equal the car's:

```
gain = (ratio − 1) × L(bench background) / L(PGU white at the eye)
ratio = 1 + L(HUD at the eye in the car) / L(background in the car)
```

CONDITION IN THE CAR	HUD AT EYE	BACKGROUND	RATIO	BENCH GAIN	ON THE BENCH
Plan minimum against a noon sky (1,500 at the eye)	1,500	10,000	1.150	0.35	yes
Design backlight against the sky (10,000 cd/m2 panel)	1,944	10,000	1.194	0.45	yes
Design backlight against sunlit asphalt	1,944	5,000	1.39	0.18	yes
This monitor through the Stage 1 combiner, against the sky	360	10,000	1.04	0.084	yes
Production AR HUD (10,000 at the eye) against the sky	10,000	10,000	2.00	2.33	no: beyond the monitor

Table 3.2. Conditions from the plan and the bench gain that reproduces each over the road monitor's white (sky) or its asphalt. The Stage 1 row is this same monitor behind a 30% combiner in a real car: the bench shows at gain 0.084 how invisible that is against a sky, which is the argument for the custom backlight. Production brightness lies beyond the monitor.

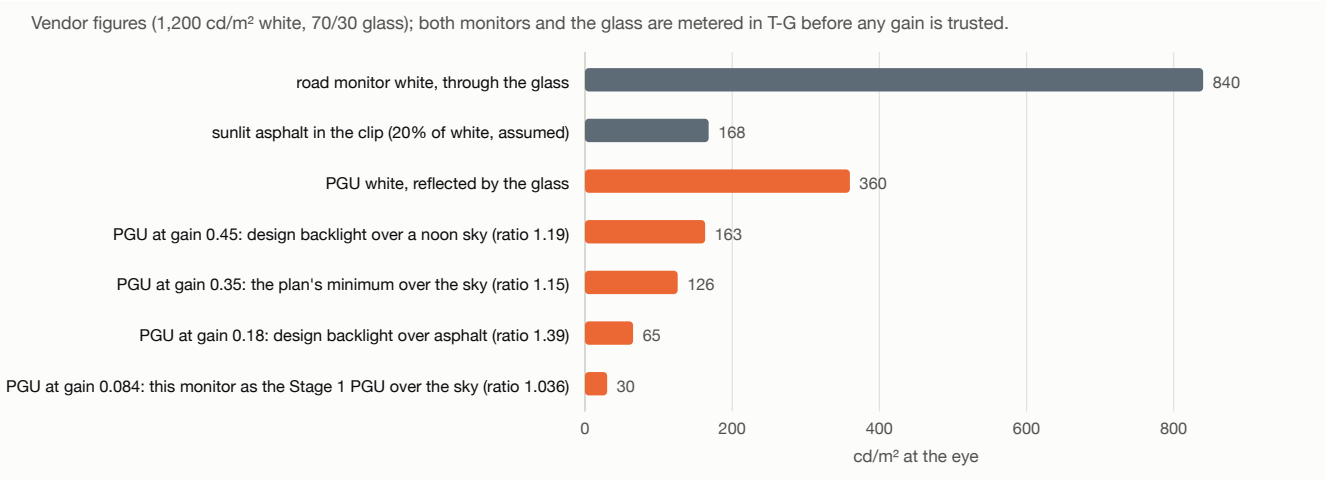


Figure 3.3. Luminance at the eye on the vendor's figures, and the PGU at the gains of Table 3.2.

Gains are applied in linear light, as the road simulator does: a drawn value c is shown as $c^{1/2.2} \times gain^{1/2.2}$ on an sRGB monitor, because the ratio is defined in luminance. The road monitor's own brightness setting is the second knob: lowering it to a metered value emulates an overcast or a dusk background. What the bench cannot do is raise the background to 10,000 cd/m²: the ratios are right, the adaptation is not, and the

judgement of legibility at noon transfers only approximately. The unlit window's glow, the monitor's black through the glass, is about 0.30 cd/m² at full brightness on a 1,200:1 panel, far under the plan's night ceiling and the simulator's finding of 50 cd/m² being too bright; the night black-level test belongs to Stage 2.

3.5 Polarisation

An IPS monitor emits linearly polarised light. Plain glass at 45° reflects s-polarised light at 9.7% and p-polarised at 0.94% (n = 1.52, refracted angle 27.7°), a ratio of 10.3; the beamsplitter coating's split between s and p is not published for the NA17, and the "30%" is for whichever light the vendor measured. With the PGU face up and the glass tilted about a left-right axis, the plane of incidence is vertical and s is the left-right polarisation. If the panel's polariser is at 45° to its edges, as many are, the reflected share is a mix; if it is left-right, the reflection is strongest; if it is up-down, the coating may reflect far less than 30%. VERIFY in T-G: meter the PGU's white in the glass; if it is well under 30% of its direct reading, a half-wave retarder film on the PGU (in the plan's BOM) rotates the polarisation to s, or the PGU is rotated 90° in its well with the picture rotated in software. The road monitor is unaffected: transmission at 45° differs between s and p by only a few percent.

3.6 The ghost

The glass's second face reflects too. For 2.0 mm glass at 45° the two reflected rays run 1.49 mm apart, which from 600 mm is 8.5′ or 16 monitor pixels, 6.0 times the plan's 1.4′ windshield ghost at 7.5 m. Its brightness relative to the main reflection depends on the second face: with the anti-reflection coating on that face about 0.8% (residual reflectance 0.5%, ASSUMED); on a bare face 15.8% for s-polarised light, 1.5% for p, 8.7% unpolarised. So the orientation of the glass matters: coating toward the PGU with the anti-reflection face toward the viewer gives a ghost 16 px away at under 1%, invisible on filled areas and faint on a line; the other way round gives up to a sixth of the line's brightness 16 px away, a visible double. The 58% ghost of the car's plain laminate is a different problem with a different cure (thick lines, filled areas, wedge glass), and the bench says nothing about it.

3.7 What the glass costs the picture

Nothing else: a 2 mm plate at 45° shifts the transmitted road picture by a millimetre and adds no distortion the homography cannot absorb; the window's 113.9 mm is well inside the glass's 222.7 mm reflected depth and 315 mm width; the eye sees the whole road monitor through the glass (section 2.3). Keep the glass clean and the hood on: a fingerprint on either face is a haze on both pictures.

3.8 The parallax variant

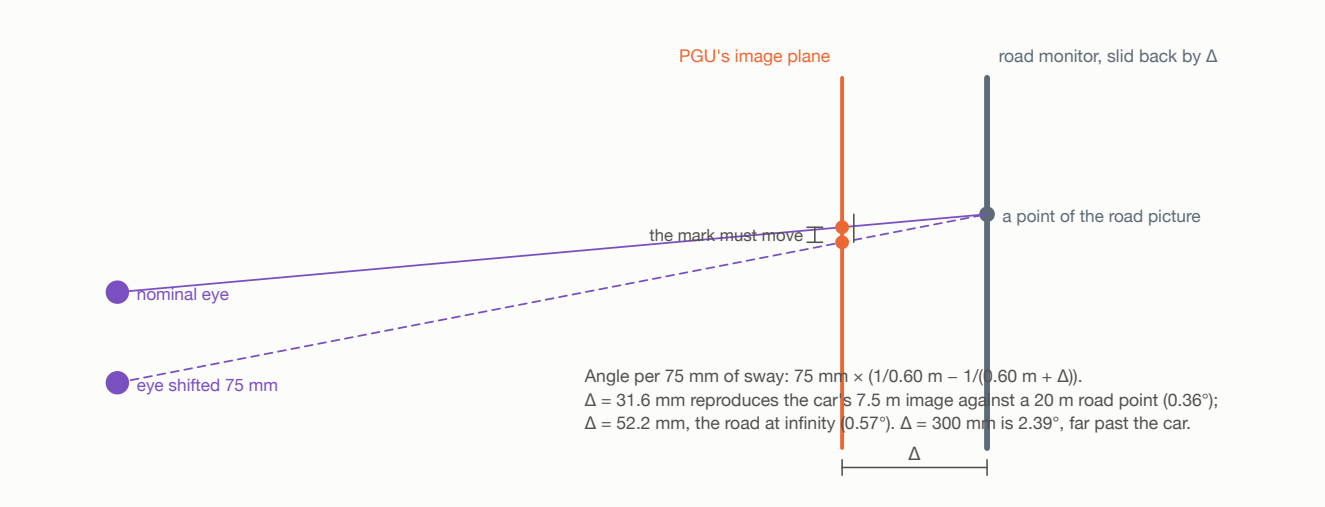


Figure 3.4. With the road monitor slid back by Δ, a sideways move of the eye slides the road behind the PGU's image. The head tracker's job is to move the mark by the same angle.

In the car the virtual image sits at 7.5 m and the road behind it at 10 to 60 m, so a head movement shifts the overlay against the road and the head tracker exists to correct it. The bench reproduces this by sliding the road monitor back a distance Δ while the PGU's image stays at 600 mm. The angle per 75 mm of sway is 75 mm × (1/0.60 m − 1/(0.60 m + Δ)); solving for Δ reproduces the car's parallax exactly:

ROAD POINT IN THE CAR (IMAGE AT 7.5 M)	Δ ON THE BENCH (MM)	ANGLE PER 75 MM OF SWAY
10 m	12.2	0.14°
20 m	31.6	0.36°
30 m	38.3	0.43°
60 m	45.2	0.50°
infinity	52.2	0.57°
(no car equivalent)	100	1.02°
(no car equivalent)	300	2.39°

Table 3.3. Δ that gives the bench the car's parallax between its 7.5 m image and a road point at the stated range. A large Δ is a stress test, not a car condition.

In this variant the mapping from a road pixel to a PGU pixel depends on the eye: `hudd` projects from the tracked eye through the road point and intersects the image plane (chapter 6). The webcam tracker's residual is measured with the camera on the rail at ±75 mm, tracker on and off, against the plan's 0.3° criterion, the test the plan assigns to the car (its §1) brought forward to the desk.

PART II

Detection

Where the road pictures come from, what each one knows about itself, and how lanes, vehicles and pedestrians are found in them and placed on the road.

---

4 Road sources · 5 Detection design

4 Road sources

A road source is a stream of frames with a camera model, a clock, and whatever it knows about the car's motion and the truth of the scene. The bench accepts four, behind one interface, chosen for what each contributes: real paint and real motion, exact truth, sharpness, or the live case.

SOURCE	FRAME (PX)	F (PX)	FIELD	HZ	CROP (PX)	SCALE	WINDOW (PX)	WHAT IT GIVES
comma2k19 segment (comma EON camera)	1164 × 874	910	65.2°	20	534	× 7.2	173	MIT; video.hevc + processed_log (CAN speed, gyro, GNSS) + global_pos
openpilot route (comma three, narrow camera)	1928 × 1208	2,648	40.0°	20	1,554	× 2.5	503	demo route used by tools/clip; logs carry modelV2, liveCalibration, carState
ZED 2i, 4 mm lens, 1080p (car camera; bench loop only)	1920 × 1080	1,144	80.0°	30	671	× 5.7	217	store page 80 degrees; datasheet 72 (conflict in the plan)
phone dashcam clip, 4K (ASSUMED 70 degrees)	3840 × 2160	2,742	70.0°	30	1,609	× 2.4	520	no log; ego-motion from a GNSS app or none
MetaDrive render (docs/arhud-road)	3840 × 2400	6,543	32.7°	60	3,840	× 1.0	1,242	rendered at the monitor's own field; truth for every lane, car and pedestrian

Table 4.1. Sources and their cameras. Focal lengths for the comma cameras are this checkout's `openpilot/common/transformations/camera.py` (`_neo_config` for the EON, `_ar_ox_config` for the comma three); the ZED's is the plan's. "Life-size crop" is the source width that fills the road monitor at 600 mm; "window" the source pixels across the plan's 10.8°.

4.1 comma2k19 segments

The dataset the plan's reference stack came from: 2,019 one-minute segments of Highway 280 recorded on a comma EON, MIT licensed, about 100 GB in ten chunks of about 10 GB on Academic Torrents, also mirrored as the Hugging Face dataset `commaai/comma2k19` VERIFY; one chunk is enough. Each segment holds `video.hevc` (1164 × 874 at 20 Hz, focal length 910 px), `processed_log/` with CAN speed in m/s and the four wheel speeds, the IMU's accelerometer and bias-corrected gyro in a forward-right-down frame, GNSS, and `global_pos/` with per-frame positions and orientations from the authors' Laika solution, plus `raw_log.bz2` readable with openpilot's `LogReader`. The frame reader in `openpilot/tools/lib/framereader.py` decodes the HEVC through ffmpeg, which on the Mac uses the hardware decoder.

What it gives the bench: real paint under real light, and the car's own speed and yaw rate at log rate for `egod`, so age compensation on the bench uses the same quantities the car will have. What it lacks: a camera height and pitch per car. The height is taken as 1.22 m, the plan's own EON-era figure ASSUMED; the pitch per segment is fitted from the vanishing point of the lane lines on a straight stretch, cross-checked against the

Laika orientations (chapter 8). The camera's 65.2° field is wider than the 32.7° shown, so the life-size crop uses only its central 534 px and the picture is soft (section 3.3); the lane and object networks see the full-resolution source.

4.2 openpilot routes from a comma three

A route recorded on a comma three carries a narrow camera of 2,648 px focal length on 1928 × 1208 (40.0° wide, 20 Hz) and, in its logs, `liveCalibration` (the camera's pitch and yaw against the road and, in newer logs, its height), `carState` (speed, yaw rate from the car's sensors), and `modelV2`: openpilot's own lane lines and lead, which serve the bench as a reference for the lane detector before any hand labelling. The demo route this checkout's `tools/clip` uses, `5beb9b58bd12b691/0000010a--a51155e496`, is public VERIFY; the repository's `tools/lib/route.py` fetches segments. Its 40.0° field is close to the shown 32.7°, so the picture needs only × 2.5 enlargement and looks sharp at the eye; the camera sits high on the windshield at the comma device's own calibration.

4.3 MetaDrive renders

The road simulator in `docs/arhud-road` already renders a road camera view from a mount on the windshield and keeps the truth of every lane boundary, vehicle and pedestrian. A new export tool in it writes a bench recording: a frame sequence at 3,840 × 2,400 rendered with exactly the field the road monitor shows (32.7° × 20.8°, a pinhole of 6,543 px), a second sequence at the ZED's resolution and field for perception, the camera pose, the true ground plane, the ego speed and yaw rate at 100 Hz, and the truth per frame as car-frame polylines and boxes. It is the only source with truth, so it is the one the perception score (§5.7) and the registration residuals (chapter 6) are measured on; its paint is synthetic, so the lane detector's score on it is a floor, not a result. The simulator's scenarios (lead braking, pedestrian crossing, worn paint, curves) become bench recordings directly.

4.4 Phone dashcam clips

A 4K clip from a phone held on a windshield mount, 30 or 60 Hz, about 70° wide ASSUMED, is the sharpest real picture the bench can show (× 2.4 enlargement) and the easiest to record on the roads the demo will use. It has no log: the camera's focal length is fitted from a chessboard once, its pitch from the vanishing point, and ego-motion either from a GNSS logging app on the same phone (speed, heading, 1 to 10 Hz) or not at all, in which case age compensation is off and the clip serves the detector and the drawing rules only.

4.5 Live camera

The same interface accepts a camera on the Mac or the Jetson (a webcam, the OAK-D Pro, the ZED 2i when it arrives) so that the bench becomes the car's perception chain with the road monitor as the scene. This is the camera-in-the-loop configuration of chapter 9; its frames carry the camera's own capture time and its own intrinsics, and a homography maps its view of the road monitor back to road-picture pixels.

4.6 The source interface

Every source yields frames with: a monotonic frame index; the source's native image; the camera model (focal length, principal point, distortion if known, image size); the mount (height, pitch, yaw, lateral offset from the car's centre); the ego sample nearest the frame (speed, yaw rate, their times) and the full ego stream for `egod`;

the truth, if any; and a display crop (the window of the source that fills the road monitor at life size, so that the pixel mapping between source and road monitor is a known scale and offset). `roadsrc` shows the crop and publishes the frame; the rest of the chain never knows which source it came from.

TIME ON THE BENCH

A recording has its own timeline. `roadsrc` replays it in real time, scheduling each frame to a vsync of the road monitor, and defines the frame's **capture time** as the moment its scan-out begins on the road monitor (the vsync plus the monitor's measured input lag). Every topic derived from the frame carries that time, so `hudd` 's age is the real age of the light the viewer saw. The ego stream is shifted onto the same timeline. Sync mode (chapter 6) uses the same clock with perception precomputed.

5 Detection design

The bench runs the plan's perception processes on the source frames: a lane network, an object detector with a tracker, and the lifting of their pixels to the car frame. Only the lifting differs from the car, because there is no stereo: it uses the recording's camera model and a static ground plane, the plan's own fallback mode, which the bench makes measurable.

5.1 Processes

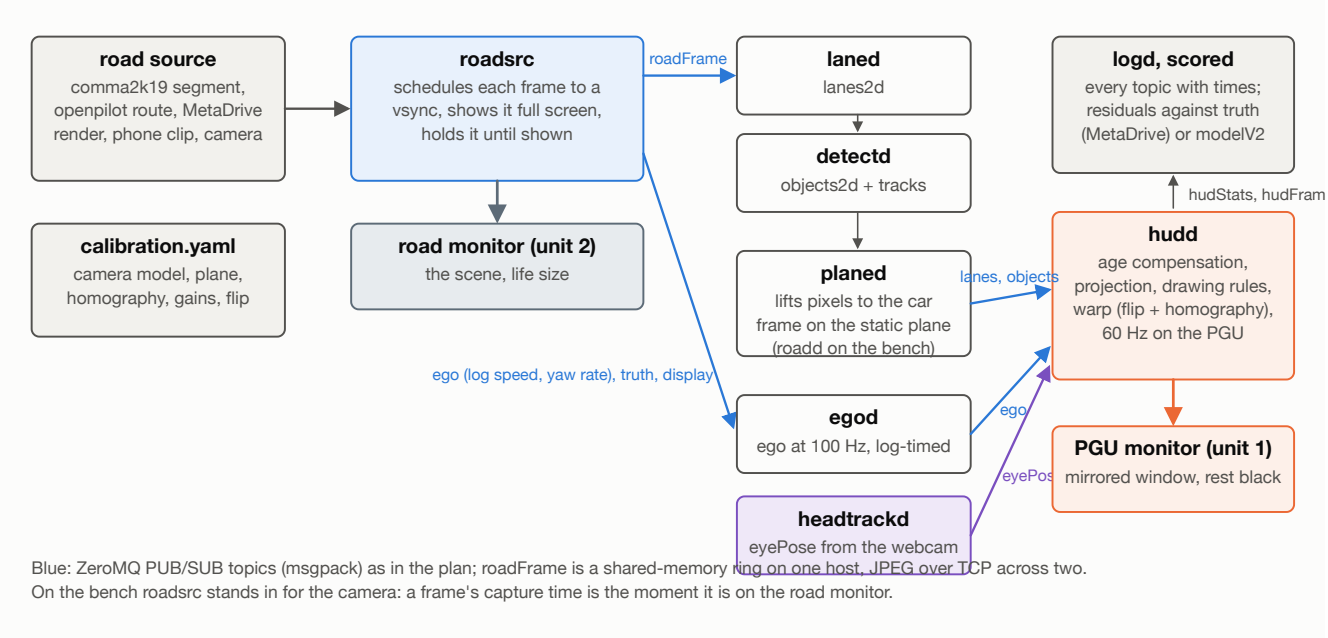


Figure 5.1. Processes and topics on the bench. Names and schemas are the plan's (§3.1, §3.2); `roadsrc` and `planed` are the bench's stand-ins for the ZED's `roadd`, and `truth` and `display` are bench-only topics.

PROCESS	ON THE BENCH	PUBLISHES
`roadsrc`	Opens a source (chapter 4), shows its life-size crop full screen on the road monitor one frame per scheduled vsync, and releases each frame to perception at its capture time. Publishes the ego stream on the bench timeline and the truth when the source has it. Draws the frame-ID strip and the timing patch (chapter 7).	`roadFrame`, `ego`, `truth`, `display`
`laned`	Lane network on the source frame. Up to four lines as image points with confidence, plus road edges if the network gives them. Same process as the car's; TensorRT on the Jetson, ONNX Runtime or Core ML on the Mac.	`lanes2d`
`detectd`	Object detector and tracker on the source frame; sign crops to the classifier when that exists. Same process as the car's.	`objects2d`
`planed`	Lifts `lanes2d` and `objects2d` to the car frame through the source's camera model and the static ground plane; smooths lane points over time (the build guide's correction); assigns ranges to tracks from the box base on the plane. It is `roadd`'s lifting code without the stereo and the plane fit, so that in the car the fit simply replaces the static plane.	`lanes`, `objects`

PROCESS	ON THE BENCH	PUBLISHES
`egod`	Republishes the source's speed and yaw rate at 100 Hz on the bench timeline, with the plan's fields (pitch and acceleration where the source has an IMU, fix quality from the source's GNSS or "simulated"). The car's `egod` fuses the F9R and the ZED's IMU into the same message.	`ego`
`headtrackd`	Webcam on the road monitor's top edge facing the viewer; face landmarks on the Mac (MediaPipe runs there, unlike on the Jetson) give the eye midpoint in the rig frame from the webcam's calibrated pose. Only the parallax variant uses it.	`eyePose`
`hudd`	The renderer, chapter 6.	`hudFrame` , `hudStats`
`logd` , `scored`	Record every topic with times; compute perception scores against truth and registration residuals from `hudd` 's own projected points against truth or the reference.	files

5.2 Frames and their path

On one host the frame travels by a shared-memory ring: `roadsrc` writes the source image once, publishes a small `roadFrame` message (index, capture time, slot, camera model id) and the networks read the slot. Across two hosts (the Jetson for perception, the Mac for the road monitor) the frame goes as JPEG over TCP at the source's rate, which costs a few milliseconds and is measured as part of the chain. The networks never see the enlarged picture on the road monitor; they see the source at its native size, as the car's networks will see the ZED's frame. For the camera loop (chapter 9) the camera's frame replaces the source image and a homography maps its detections into source pixels before lifting, so `planed` and everything after it are unchanged.

5.3 The lane network

The plan's order stands: Ultra Fast Lane Detection v2 first (MIT, CULane weights, 1,600 × 320 input crop, ONNX export in its repository), CLRNet or CLRerNet second (Apache, higher F1, slower), CondLaneNet and YOLOP as alternatives. The bench's contribution is the measured choice: each candidate runs on the same 300 frames with truth (MetaDrive) and on the same comma three segments against `modelV2` , and the score of §5.7 decides. Inputs are the source frame resized to the network's crop with the horizon placed where the network's training data had it; outputs are row-anchored image points with a confidence per line. On the Mac the networks run through ONNX Runtime's Core ML provider or PyTorch on Metal; on the Jetson through TensorRT FP16 from the same ONNX file, so the bench score and the car's networks are the same weights.

5.4 The detector and the tracker

YOLO11s at 640 px, FP16, the plan's eight COCO classes; exported once to ONNX (and to Core ML for the Mac). ByteTrack on the CPU for identities and the per-track velocity the renderer advances by. The alternatives the plan names (RF-DETR Nano, DashCamNet) are tried only if YOLO11s misses the recall criterion on the bench. Signs: the crop classifier for US speed-limit values is a stretch goal here as in the plan; the bench has the frames to train it on later.

5.5 Lifting to the car frame

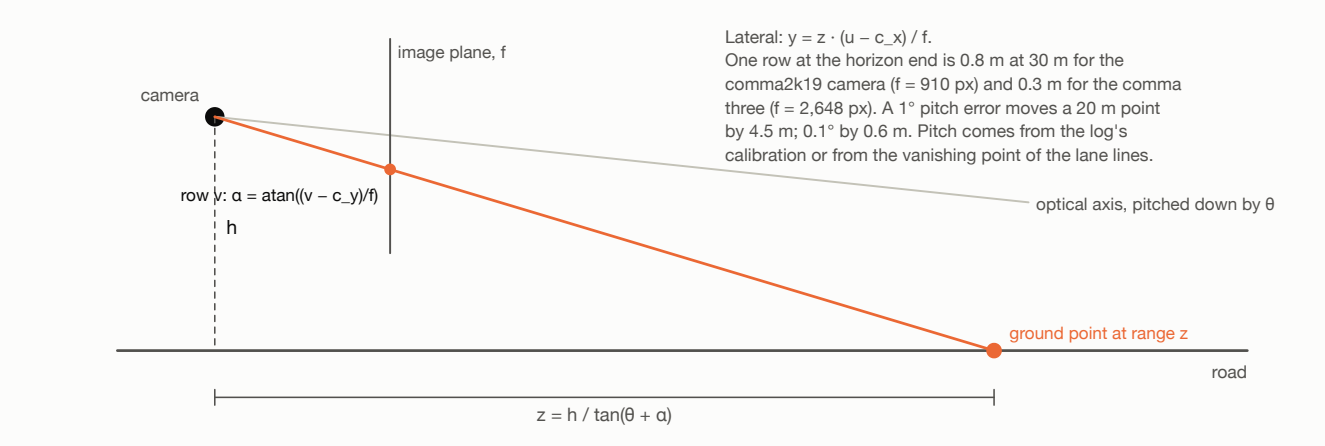


Figure 5.2. A pixel row and the static plane. Pitch and height come from the source; one row near the horizon is a large step in range.

For a pixel (u, v) in a camera of focal length f, principal point (c_x, c_y), height h and pitch θ (down positive): the ray's depression below the axis is α = atan((v − c_y)/f), its range along the ground z = h / tan(θ + α), and its lateral position y = z · (u − c_x)/f (to first order; the code intersects the ray with the plane exactly). Lane points are lifted row by row and the far end is cut where the row-to-range step exceeds a metre; a vehicle's range is its box base lifted the same way; a pedestrian's likewise. In the car, `roadd` replaces h and θ by the fitted plane every frame and the stereo range inside 35 m; the lifting function is the same.

CAMERA	ROWS BELOW THE HORIZON AT 10 M	M PER ROW	AT 20 M	M PER ROW	AT 30 M	M PER ROW
comma2k19 (f = 910 px)	111.0	0.1	55.5	0.4	37.0	0.8
comma three (f = 2,648 px)	323.1	0.0	161.5	0.1	107.7	0.3

Table 5.1. Range per pixel row near the horizon for the two comma cameras at 1.22 m. A 1° pitch error moves a 20 m point by 4.5 m (the plan's 4.3 m at its camera height), 0.1° by 0.6 m: the pitch calibration of chapter 8 must be good to a tenth of a degree.

Lane points are smoothed over time before publishing, per line, with a short exponential filter keyed to the point's range, because the simulator showed raw per-frame points making the drawn line wobble; the filter's constant is a config value the bench tunes by eye and by the residual's variance. Objects are not smoothed beyond the tracker's own state.

5.6 Ego-motion

comma2k19 gives CAN speed and the IMU gyro; a comma three route gives `carState`; MetaDrive gives the truth; a phone clip gives GNSS speed or nothing. `egod` publishes whichever exists at 100 Hz by interpolation, flags the source, and sets fix quality so that `hudd`'s rules (prediction off without a fix, speed dash after 1 s) can be exercised by cutting the stream on a key press, the F7 scenario of the road simulator on real data. The plan's GNSS-latency parameter is measured in the car; on the bench the recording's own timestamps are exact.

5.7 Scoring

Against MetaDrive truth, the plan's gate G1 as written: ego lane found in ≥ 95% of frames; lateral error ≤ 0.15 m at 10 m, 0.30 at 20, 0.50 at 30 (seen from the camera these are 0.86°, 0.86° and 0.95°, or 98.4, 98.4 and 109.4 px on the road monitor); vehicles in the ego lane within 50 m recalled ≥ 90%; pedestrians within 30 m ≥ 85%; range within 10% of truth. Against comma three routes, agreement with `modelV2` 's lane lines at the same ranges, reported but not gated, since the reference is itself a model. The scorer works in the source's image, by lifting the truth through the same camera model and comparing in the car frame, so perception error and lifting error can be separated by also scoring in pixels.

5.8 Compute

JOB PER FRAME ON THE MACBOOK PRO M4	MIN (MS)	MAX (MS)
Lane network, UFLDv2 (CULane, 1600 x 320 input), ORT CoreML or MPS	8	15
Object detector, YOLO11s at 640 px, CoreML FP16	12	25
ByteTrack association (CPU)	1	2
Lifting, tracking bookkeeping, publish	1	2
hudd draw (two passes, 1,237 x 617 window)	2	4
Serial total on the M4 (assumed)	**24**	**48**

Table 5.2. Assumed per-frame times on the M4 **ASSUMED**; measured in T-D and entered here. The plan's Jetson table (§5.6) is 10 to 25 ms for inference alone. At the assumed Mac times the chain runs at 21 to 42 Hz, enough for the 20 Hz comma sources and the 30 Hz ZED.

Perception runs at the source's frame rate, never faster; `hudd` runs at 60 Hz regardless and draws from the newest message, which is what makes age compensation necessary even on the bench (a 20 Hz source leaves a mark up to 50 ms stale before any latency). The Mac has no CUDA; the networks' Mac path exists for the bench and the renderer bench only, and the Jetson's TensorRT path is measured when the Jetson is back (B2 of the build guide must come first: reflash to JetPack 6.2).

PART III

Display

How a point on the road becomes a lit pixel on the PGU whose reflection covers it; when that pixel lights; how the chain is calibrated; and the variant with a camera in the loop.

---

6 The renderer on the bench · 7 Clocks and latency · 8 Calibration · 9 Camera in the loop

6 The renderer on the bench

`hudd` is written once, for the car. On the bench it runs with a different set of matrices in its projection chain and a different warp mesh, both read from the calibration file, and with the drawing rules of the plan unchanged. This chapter gives those matrices and the two modes in which the bench runs it.

6.1 The projection chain

The plan's chain is D · K_{eye} · view_from_eye · eye_from_car: a car-frame point goes to the eye's frame, is projected as a pinhole onto the virtual plane, and is warped by a two-dimensional mesh onto the panel. On the bench the "virtual plane" is the road monitor's face, which shows the recording's camera picture; a car-frame point appears on it where the recording's camera projected it. So the bench's eye is placed at the recording's camera, and the chain becomes:

```
eye_from_car   = camera_from_car of the source (height h, pitch θ, yaw, lateral offset)
view_from_eye  = identity (look along the camera's axis)
K_eye          = S · K_source, with S the scale and offset of the life-size crop onto the road monitor
D              = H_PR · F : the mirror flip F, then the calibrated homography from road-monitor pixels to PGU pixels
```

Every matrix is a config value. In the car, eye_from_car is the nominal eye plus the tracker's offset, K_{eye} the virtual plane's pinhole and D the mesh from the photographed grid; the code path is identical, which is the point. The bench's D is a 3 × 3 homography rather than a mesh because a flat reflection needs no more; `hudd` carries the general mesh and fills it from the homography.

The window: the plan's 10.8° × 5.4° is drawn as 1,242 × 621 px on the road monitor's plane, its centre the plan's 5° look-down below the picture's horizon row ($c_y - f \cdot \tan\theta$ for a camera pitched down by θ, the row where the lane lines vanish). `roadsrc` chooses the life-size crop's vertical offset so that this row lands on the road monitor's centre row, which is also the glass's centre; the window is then centred in the glass's reflected area and tied to the horizon at once. Only that region is drawn on the PGU; the rest stays black. The two-pass structure of B6.4 stands: content into an off-screen texture at twice the window's resolution, then the warp pass through D.

6.2 Age compensation on the bench

As in the plan: for each point captured at t_c, age = $t_{photon} - t_c$, where t_{photon} is the predicted time the drawn frame lights the PGU (now, plus the measured render and scan-out time of chapter 7); speed and yaw rate from `ego` are integrated over the age to give s and ψ, and the point is moved by $p_{now} = R_z(-\psi)\cdot(p_{capture} - [s, 0, 0])$, tracks additionally by their own velocity. The bench differs in one respect that makes the test honest: the road monitor shows the frame of time t_c, not the road at t_{photon}. So a correctly compensated mark is drawn where the road *will be*, ahead of the picture, and looks wrong against the picture; a mark compensated with the wrong sign looks doubly wrong. Registration against the picture is therefore measured either in sync mode (section 6.4) or, in live mode, with `hudd`'s compensation target set to the picture's own time, a config switch that the plan's B6.5 sign test also needs (it draws onto the camera image). Live mode's own measurement is the chain latency and the residual of the compensated mark against the truth at t_{photon}, which MetaDrive sources provide.

6.3 Drawing rules

Unchanged from the plan's §6 and the build guide's B6.4: ribbon between the ego lines, lane lines faded by confidence with three frames before appearance, lead chevron within 80 m in the ribbon, hazards after three detections, the fixed speed block, nothing below the window's bottom edge or above the horizon plus 2°, the stale cut-offs, clutter mode on a key. Two bench additions: a **gain** (section 3.4) that scales every drawn value to reproduce a chosen contrast condition, and a **blank-ribbon** switch (the simulator's) for registration checks. Line widths are in window pixels and scale with the window's pixel count, so a 6 px line on the 5.5-inch crop is 115/87 × 6 ≈ 8 px here.

6.4 The two sync modes

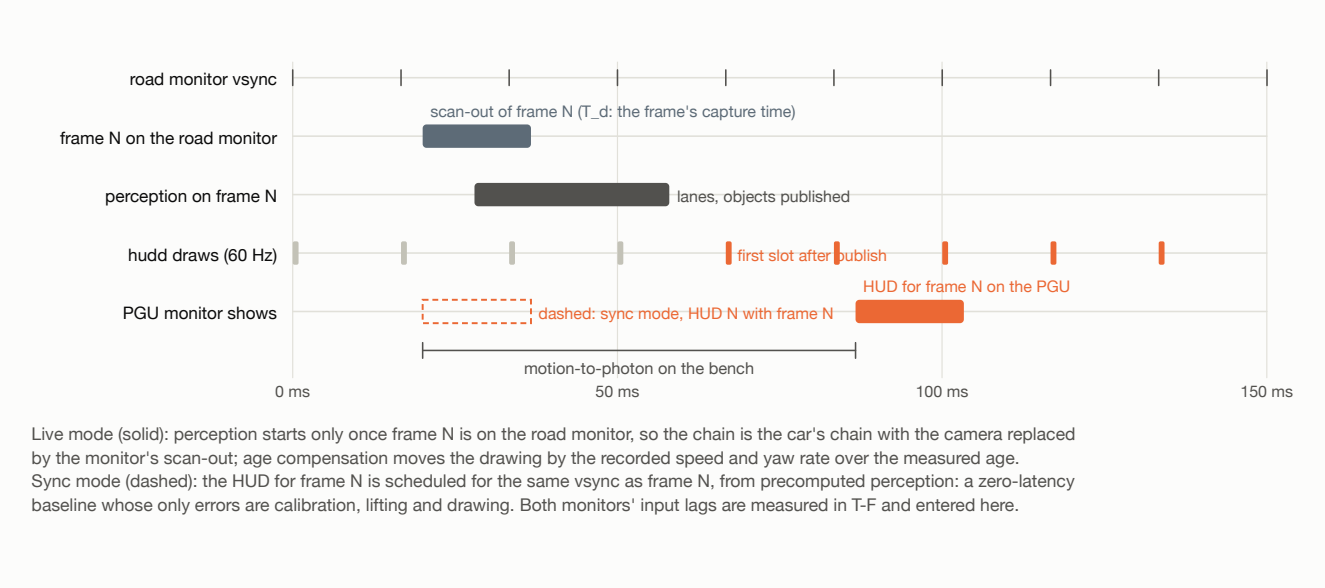


Figure 6.1. One frame through the bench in live mode (solid) and sync mode (dashed).

MODE	WHAT HAPPENS	WHAT IT MEASURES
Sync, truth-fed	`hudd` subscribes to `truth` in place of `lanes` and `objects` (MetaDrive sources only). `roadsrc` and `hudd` share a schedule: the HUD drawing for frame N is presented at the same vsync as frame N on the road monitor, offset by the difference of the two monitors' input lags. Age is zero by construction; compensation is off.	Calibration, projection and drawing alone: the residual between a drawn line and the painted line's truth is the display chain's error, with nothing else in it.
Sync, perception-fed	As above, but perception runs ahead of time (or offline) on the source frames and its results are keyed by frame index; `hudd` draws from them.	The whole chain without latency: perception, lifting and display together, reported against the §5.7 angles (0.86° at 10 m, 0.86° at 20 m, 0.95° at 30 m).
Live	Frame N is released to perception at its capture time (when its scan-out begins on the road monitor); `hudd` draws from the newest message at 60 Hz with compensation to t_{photon} using the recording's ego stream.	The latency chain, the prediction's sign and magnitude, frame drops, and the compensated residual against truth at t_{photon}. B6's gate G2.

MODE	WHAT HAPPENS	WHAT IT MEASURES
Live, fixed delay	Live with an added delay of 200 ms before publication (B6.5).	The sign test on a curve: the compensated mark lands on the paint's future position; the uncompensated lags; a wrong sign doubles the error.

6.5 The parallax variant's projection

When the road monitor is slid back by Δ, the road picture is a plane at 0.60 m + Δ and the PGU's image a plane at 0.60 m. A road point P_{road} (where the recording's camera put it, scaled onto the road monitor) must appear to the tracked eye E to coincide with the mark: the mark goes where the line from E through P_{road} crosses the image plane, and then through F and H_{PR} to the PGU. `hudd` does this with the plan's own machinery: eye_from_car is the tracked eye in the rig frame, K_{eye} the pinhole of the image plane as seen from that eye, and the road point is first mapped to its rig-frame position on the slid-back monitor. With the tracker off, E is the nominal eye and the mark errs by exactly the parallax of Table 3.3, which is the test.

6.6 Output and the audience

raylib (OpenGL 3.3 core) full screen on the PGU at native 3,840 × 2,400, vsync on, one process per monitor so that each full-screen window owns its display. On macOS the window must be created at physical resolution (the high-DPI flag) or the system will render at 3,840/2 and scale. `hudFrame` (the window, downscaled) and `hudStats` (age per topic as drawn, render time, dropped frames, the last residual) are published as in the plan; an audience view, when wanted, is the Mac's own screen: road picture with perception overlaid, the PGU frame, the live numbers, as the road simulator's right panel lays them out.

7 Clocks and latency

The plan's motion-to-photon budget is 37 to 99 ms with a limit of 90. The bench reproduces the chain with the camera replaced by a monitor and measures it with two photodiodes, without a car and without the ZED.

7.1 One clock, or two

With everything on the Mac there is one monotonic clock and every topic's capture and publish times are on it. When the Jetson takes the PGU and the perception, the Mac keeps the road monitor and `roadsrc`; the two hosts run chrony over the Ethernet switch, which holds them within a millisecond on a quiet LAN VERIFY, and the frame's capture time crosses with the frame. A millisecond is 1% of the budget; the photodiode measurement does not depend on it at all, because it reads light on both monitors with one Pico.

7.2 What "capture time" means on the bench

In the car a frame's capture time is the ZED's exposure midpoint. On the bench it is the moment the frame's light leaves the road monitor: the vsync at which `roadsrc` presented it plus the road monitor's input lag (vsync to the start of scan-out), measured in T-F. `roadsrc` knows the vsync from the display link's timestamp and holds the frame back from perception until that capture time has passed, so perception cannot start before the light exists, as in the car. The ego stream is shifted so that the recording's own timestamps line up with these capture times.

7.3 The bench chain

STAGE	MIN (MS)	MAX (MS)
Road frame on the road monitor (its scan-out and input lag, hidden by scheduling)	0	0
Perception on the host (lanes, objects, tracking)	15	40
Publish and pick-up by hudd	1	3
hudd render (60 Hz, vsync)	8	16
PGU monitor scan-out and input lag	10	30
Total, road photon to HUD photon	**34**	**89**
Plan's car chain, for comparison (camera exposure to panel)	37	99

Table 7.1. The bench chain by stage, road photon to HUD photon. Monitor lags and Mac inference times are ASSUMED until T-F and T-D measure them. The road monitor's own scan-out is excluded because the capture time is defined at its start.

A 20 Hz source adds up to 50 ms of staleness before this chain, which is the car's situation at 20 Hz too; the plan runs the camera topics at 30 Hz if the GPU allows. On the road, 1.5, 2.7 and 6.0 m pass at 30 m/s in 50, 90 and 200 ms, and a lane point 10 m ahead seen from 1.22 m up moves by 1.21°, 2.53° and 10.01° in the image; that is what compensation removes and the bench shows with real speed and yaw rate.

7.4 Measuring it

Frame-ID strip. `roadsrc` draws eight small squares at a corner of the road monitor encoding the frame index modulo 256; the loop camera (chapter 9) and the measuring camera decode it, so any photograph or video of the rig knows which frame it sees. It costs nothing and resolves every "which frame was that" question.

Timing patches and the photodiode pair. On a key press, or at random intervals of 0.5 to 1.5 s, `roadsrc` turns a 60 × 60 px patch at the road monitor's corner white for one frame and publishes a flag with that frame's capture time; `hudd` turns a 60 × 60 px patch on the PGU white in the first frame it draws after receiving the flag, through the normal path (the flag travels as a field of `roadFrame` through `laned`, `planed` and the bus, so it takes the full chain). B6's Pico probe gains a second photodiode: one taped over each patch, shielded. The Pico timestamps both rising edges in microseconds and prints the difference; 200 samples give the median and the 95th percentile. The road monitor's own input lag is read the same way against the Pico receiving the vsync timestamp over serial, and the PGU's by sending a byte that `hudd` answers with the patch, which is also the renderer-plus-monitor figure B6.6 wants.

Phone slow motion for a quick look: a phone at 240 fps filming the glass sees both patches (the road one through it, the PGU one in it); the frame count between the two edges times 4.2 ms is the chain to the nearest 4 ms. Good enough to catch a stacked frame or a wrong vsync setting in a minute.

7.5 Input lag and vsync on macOS

Each monitor's lag is unknown VERIFY; portable panels with scalers range from one to three frames. The vsync path on macOS can add a frame of its own if the window is composited rather than full screen, or if triple buffering is on; `hudd` is checked for exactly one frame between draw and scan-out by the byte-to-patch test, and the mode that gives it (full-screen exclusive, vsync on, no compositor) is recorded in the config. The road monitor's lag enters the capture-time definition; the PGU's enters t_{photon} for compensation and the sync-mode offset.

7.6 Two hosts

When perception moves to the Jetson, the frame crosses the LAN as JPEG (a 1164 × 874 frame at quality 90 is about 150 kB, about 1.5 ms on gigabit plus encode and decode), the topics cross as they already do over TCP instead of IPC, and the photodiode pair still reads the chain end to end. The Jetson's own HDMI lag and TensorRT times replace the Mac's in Table 7.1.

8 Calibration

Seven calibrations, each with a procedure, a tolerance and the config field it fills. All are stored in `config/calibration.yaml` (appendix B) with the date and the measurement that produced them; all are repeated after the rig is moved.

8.1 Monitor geometry

Active width and height by rule, both units, to 0.5 mm; pixel pitch = width / 3,840. Fills `monitors.*.pitch_mm`. Check that the two units agree within 0.3%; if not, the homography absorbs the difference but the life-size distance is computed from the road monitor's pitch.

8.2 Coplanarity

Section 3.1's parallax check: two photographs 100 mm apart from the eye point, crosses at the window's centre and at its four corners on both monitors, separation ≤ 5 px for every cross. Corner crosses separate from tilt as well as from offset, so a residual that grows toward one side is a tilt of the road monitor or the glass, fixed by shimming. Fills nothing; it is a precondition of 8.3.

8.3 The mirror homography H_{PR}

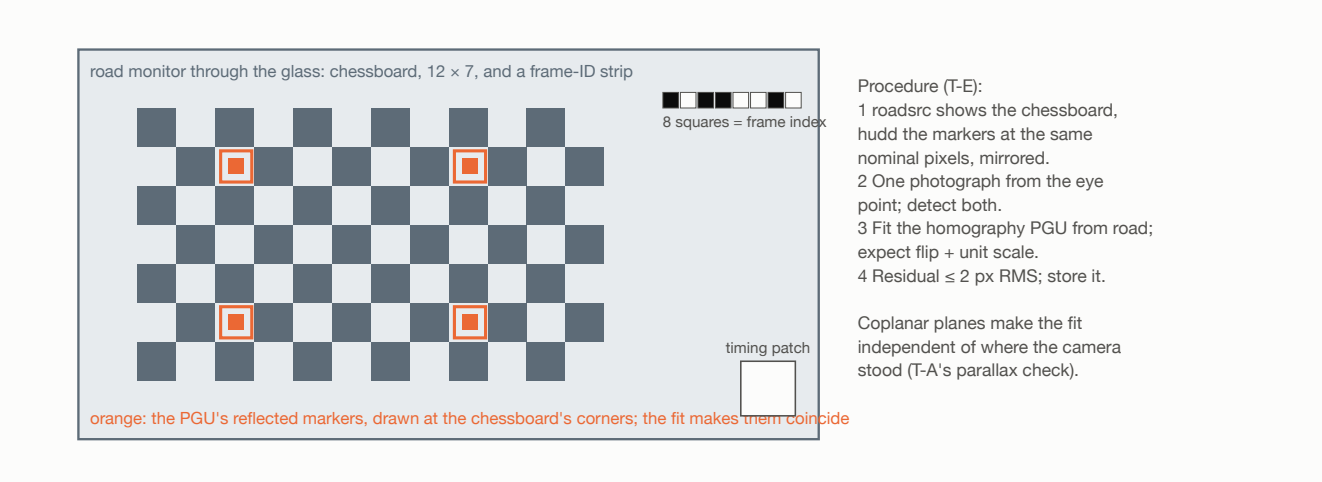


Figure 8.1. The measuring camera at the eye sees the road monitor's chessboard through the glass and the PGU's markers in it; one photograph fits the homography between them.

1. `tools/calib_rig.py` shows a 12 × 7 chessboard filling the window region on the road monitor and, on the PGU, ArUco markers at the pixels that a perfect rig (flip at unit scale) would reflect onto the chessboard's corners.
2. One photograph from the eye point with the measuring camera; the tool detects the chessboard corners and the markers in it.
3. It fits the homography from road-monitor pixels to PGU pixels that makes the markers' reflected positions coincide with the corners, and reports the RMS and maximum residual in road-monitor pixels.
   - ☐ RMS ≤ 2 px (0.017°), maximum ≤ 4 px

4 It writes `warp.homography_pr` and `warp.flip`; `hudd` reloads and shows the chessboard itself, mirrored through the new warp; a second photograph should show the two chessboards as one.

- ☐ Second photograph: no visible doubling

Because the planes coincide, the result does not depend on where the camera stood, which 8.2 guaranteed. The homography is close to the flip; a scale far from 1 means the pitches of 8.1 disagree, a shear means the glass is not square to the monitors.

8.4 The source camera and its pitch

Focal length and principal point: from the camera config for the comma sources, from the render for MetaDrive, from a chessboard sequence for a phone (OpenCV's calibrator). Height: 1.22 m **ASSUMED** for comma2k19, `liveCalibration`'s value for routes that have it, the render's for MetaDrive, a tape measure for the phone. Pitch, per recording: `tools/calib_source.py` finds the vanishing point of the two ego lane lines on straight frames (the lane detector's own output, median over 100 frames), and pitch = atan(($v_{vanish} - c_y$)/f); yaw likewise from the horizontal coordinate. Cross-check against `liveCalibration` or the Laika orientations where they exist. Tolerance 0.1°, because 0.1° moves a 20 m point by 0.6 m (Table 5.1). Fills `sources.`*`name`*`.camera` and `mount`.

8.5 Photometry

With the plan's spot meter (or a camera in manual mode as a relative meter, calibrated against one metered patch): each monitor's white direct, at the brightness settings to be used; the PGU's white in the glass from the eye point; the road monitor's white through the glass; the PGU's black in the glass. From these: the glass's actual reflection and transmission for each monitor's polarisation, the gains of Table 3.2 recomputed from measured rather than vendor values, and the unlit window's level. Then the ghost: a 2 px bright line on the PGU photographed in the glass at high magnification, the second image's offset in pixels and its brightness ratio, for both orientations of the glass; the orientation with the fainter ghost is kept and recorded. Fills `photometry.*`.

8.6 Latency

Section 7.4: the road monitor's lag, the PGU's lag, and the chain, each as median and 95th percentile over 200 samples. Fills `timing.lag_road_ms`, `timing.lag_pgu_ms`, `timing.render_ms`; the chain itself is a result, not a config value.

8.7 The head tracker

Webcam intrinsics from a chessboard; its pose in the rig frame from a photograph of a chessboard laid on the PGU's well at a known position; then the eye midpoint reported by the landmarks for the measuring camera placed at five known rail positions (0, ±75, ±150 mm). The residual of the reported position against the rail's reading, median and worst, is the tracker's accuracy; a scale error from a wrong focal length shows as a slope. Tolerance: 5 mm at the eye point, which is 0.024° of parallax in the Δ = 31.6 mm variant. Fills `tracker.*`.

8.8 The loop camera

Chapter 9: intrinsics from a chessboard; the homography from its image of the road monitor (showing a chessboard) to road-monitor pixels, residual ≤ 2 px; then the composite mapping to source pixels through the life-size crop. Fills `loop.*`.

9 Camera in the loop

The direct path hands perception the source's own pixels. The loop path puts a real camera between the road monitor and perception, so that exposure, readout, focus, the camera's transfer and its driver are in the chain, as they will be in the car. It is the last bench stage and the first in which the Jetson's camera code runs unchanged.

9.1 Placement

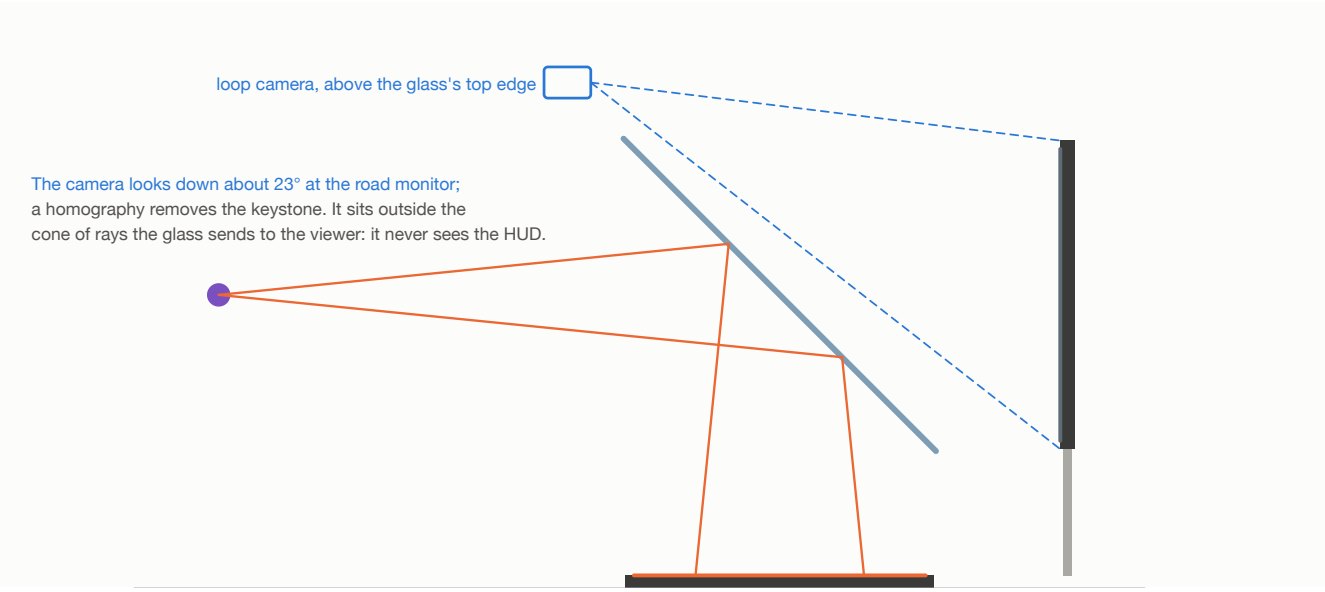


Figure 9.1. The loop camera above and behind the glass's top edge, outside the reflection's path, looking down at the road monitor.

The camera must see the road monitor and not the PGU's reflection, which means it sits outside the cone of rays the glass sends toward the viewer: above the glass's top edge, tilted down about 15° at the road monitor's centre. The car's camera has the same relation to the windshield: it looks through the glass that reflects the HUD, and the reflection goes elsewhere. The downward tilt keystones the picture; the homography of 8.8 removes it, and the camera's field is chosen (lens or crop) so the road monitor fills most of the frame, since source pixels per camera pixel set the detector's effective resolution.

9.2 A camera looking at a monitor

- **Refresh beat.** The monitor refreshes at 60 Hz; an exposure that is not a whole number of 1/60 s records a brightness band that rolls through the frames. Exposure 1/60 s or 1/30 s, fixed, gain to suit; the hood keeps room light out so the exposure can be long.
- **Rolling shutter against scan-out.** A rolling-shutter camera (the ZED 2i) reading out over a monitor's own top-to-bottom scan can capture the seam between two frames. At 1/60 s exposure the seam is a soft blend; if it shows as a tear, the capture is phase-locked to the vsync by starting it from `roadsrc`'s display timestamp. The OAK-D Pro's mono pair is global shutter and sees none of this.
- **Moiré.** The monitor's pixel grid at 91.7 µm, seen from about 0.5 m by a camera whose pixel covers several monitor pixels, is below the Nyquist limit and shows no moiré; if a sharp lens and a close mount bring it up,

a slight defocus removes it.

- **Focus and depth.** The ZED's fixed focus is set for distance; at 0.5 m it is a little soft, acceptable for the networks and marked as a bench limitation. Its stereo depth sees a plane at 0.5 m, so `roadd`'s plane fit returns the monitor's plane, not the road's: on the bench `roadd` runs with the fit disabled and the static plane of the source, exactly as `planed` does, which is the plan's fallback mode.
- **Colour and gamma.** The camera sees the monitor's sRGB rendering of the clip, not the scene; brightness and white balance are set once and left.

9.3 Mapping back to the source

The camera's detections are in camera pixels. `planed` needs source pixels, because the camera model and the plane are the source's. The composite mapping is: camera pixel to road-monitor pixel (the homography of 8.8) to source pixel (the inverse of the life-size crop's scale and offset). It is one 3 × 3 matrix, applied to every lane point and box corner before lifting; everything downstream is untouched. The frame-ID strip, decoded in the camera's image, tells `roadd` which source frame it is seeing, so that its capture time can be compared with `roadsrc`'s display time: the difference is the camera's own latency, a number the plan's budget wants and the direct path cannot give.

9.4 What the loop adds to the measurements

- Camera exposure, readout and transfer in the latency chain, by the same photodiode pair: the patch on the road monitor is now seen by the camera, and the flag is raised by `roadd`'s brightness watch on that image region, exactly B6.6's probe mode without the LED.
- The networks' behaviour on a camera's rendering of the picture rather than on the picture: noise, slight blur, the camera's colour. The lane residual through the camera is compared with the direct path's on the same frames; the difference is what the camera costs.
- The ZED SDK's capture code, its timestamps and its SVO recording on the Jetson, before a car exists.

With a live camera the bench can also show, as a demonstration, the whole car chain: the road monitor plays a drive, the camera watches it, the Jetson detects, the glass overlays. Nothing in that chain is simulated except the road.

PART IV

Build

The software layout the bench starts and the car inherits, the bring-up stages with their gates, the risks, and the bill of materials.

---

10 Software layout · 11 Bring-up stages and gates · 12 Risks and bill of materials · Appendices

10 Software layout

The plan calls for a standalone `hud/` package in the repository with no openpilot build dependency. The bench is where it is started. Everything below is the car's layout; the bench-only parts are marked.

10.1 The package

```
hud/
  pyproject.toml          numpy, opencv-python, pyzmq, msgpack, pyyaml, raylib (pyray), onnxruntime (CoreML on the Mac),
                        ultralytics (export only), av or the openpilot framereader for HEVC; TensorRT bindings on the Jetson
  hud/common/
    clock.py              monotonic time, display-link timestamps (macOS CVDisplayLink / Linux DRM vblank)
    bus.py                ZeroMQ PUB/SUB, msgpack maps, the plan's envelope (capture time, publish time, sequence)
    schema.py             one dataclass per topic; empty messages are still published
    frames.py             shared-memory frame ring on one host; JPEG over TCP across hosts
    calib.py              calibration.yaml loader with the date and provenance of every field
    geometry.py           camera models, plane intersection, homographies, the rig frame
  hud/roadsrc/            BENCH ONLY: sources/{comma2k19,oproute,metadrive,video,camera}.py, display.py (full-screen
player,
                        vsync scheduling, life-size crop, frame-ID strip, timing patch), the ego and truth republishers
  hud/laned/              lane network: ONNX Runtime or Core ML on the Mac, TensorRT on the Jetson, same ONNX file
  hud/detectd/            YOLO11s + ByteTrack; sign crops (later)
  hud/planed/             BENCH: lifting on the static plane; becomes part of roadd in the car
  hud/roadd/              CAR (and the loop): ZED capture, depth, plane fit, the same lifting; SVO record and replay
  hud/egod/               bench: log republisher; car: F9R + IMU fusion
  hud/headtrackd/           webcam + landmarks on the Mac; OAK-D Pro on-device in the car
  hud/hudd/               draw.py (rules), project.py (the chain), warp.py (mesh from a homography or a lookup table),
                        compensate.py, stats.py; one raylib window on the PGU
  hud/logd/               topic recorder; replay at original timing
  hud/scored/             perception score (§5.7), registration residuals, the bench's result tables
  hud/tools/
    testcard.py           the F, the arrow, the grid; sets the flip
    calib_rig.py          chessboard + markers -> H_PR (8.3)
    calib_source.py         vanishing point -> pitch and yaw (8.4)
    calib_cam.py          chessboard intrinsics; loop homography (8.8); tracker pose (8.7)
    photometry.py           the metering sheet and the gain recomputation (8.5)
    latency.py            Pico probe driver: patches, flags, the statistics (7.4)
    synth.py              B6.3's synthetic topics
    export_metadrive.py     lives in docs/arhud-road; writes a bench recording with truth
  config/
    bench.yaml              which monitor is which, source, mode, gain, window placement
    calibration.yaml        appendix B
  models/                 ONNX and Core ML exports (not committed)
```

10.2 Topics

TOPIC	RATE	FIELDS	WHERE
roadFrame	source rate	frame index, capture time, shared-memory slot or JPEG, camera model id, source name, flag for the timing patch	bench, loop

TOPIC	RATE	FIELDS	WHERE
`display`	source rate	frame index, scheduled vsync, measured lag, the crop in source pixels	bench
`truth`	source rate	car-frame polylines per painted line, boxes per object with class and velocity, the true plane; capture time	bench (MetaDrive)
`lanes2d`, `objects2d`	source rate	as the plan's §3.1	both
`lanes`, `objects`, `ego`, `eyePose`, `hudStats`, `hudFrame`	as the plan	as the plan's §3.2, unchanged	both
`gnss`, `health`, `sunWatch`	—	not published on the bench; `hudd` treats their absence as the plan's stale rules do	car

10.3 Running it

```
python -m hud.tools.testcard --monitor pgu                    # T-A: set the flip
python -m hud.roadsrc --source comma2k19:/data/c2k19/Chunk_1/.../3 --monitor road
python -m hud.laned & python -m hud.detectd & python -m hud.planed & python -m hud.egod &
python -m hud.hudd --monitor pgu --mode live --gain 0.45
python -m hud.tools.latency --port /dev/tty.usbmodem*          # 200 samples, prints median / p95
python -m hud.scored --log runs/2026-10-xx.log --truth         # the tables of chapter 11
```

A supervisor script starts the set, restarts a crashed process after 1 s and marks its topics stale, as the plan requires. On macOS each full-screen window is created on its own display by name, at physical resolution; the two monitors are identified in `bench.yaml` by their EDID names, since macOS may renumber them.

10.4 Environments

A new venv, `~/.venvs/arhud-hud` (Python 3.12), separate from the CAD and MetaDrive venvs: the MetaDrive one is pinned to 3.11 and a GitHub commit for the simulator and must not be disturbed; the export tool runs there and writes files the bench venv reads. On the Jetson, after the reflash to JetPack 6.2 (build guide B2), the same package with TensorRT and the ZED SDK; the Mac's Core ML path is not installed there. Model exports (ONNX, Core ML) are produced by a script and not committed.

11 Bring-up stages and gates

Nine stages, each with what it needs, what it produces and the measurement that closes it. They are ordered so that every measurement has only one new thing in it. Together they close the build guide's gate G2 and bring forward parts of G1, G3 and the head-motion criterion of G4.

STAGE	BUILDS	NEEDS · PRODUCES	GATE
T-A	The rig	Both monitors, the glass, the enclosure, the slide, the eye mount (chapter 2). Produces the measured pitches, the flip, the glass orientation, the coplanar planes.	Test card reads correctly; parallax check ≤ 5 px over 100 mm.
T-B	The package skeleton and synthetic hudd	Bus, schemas, clock, config loader, `hudd` drawing from `synth.py` on the PGU (B6.3, B6.4). Produces a renderer that draws the plan's content at 60 Hz with the warp.	60 fps, no drops, 10 min, from synthetic topics; the window at 1,242 × 621 px where the glass reflects it.
T-C	Sources and the player	One comma2k19 chunk, the openpilot demo route, the MetaDrive export, a phone clip; `roadsrc` showing each at life size with the ego stream republished. **DECISION**: the eye distance (Table 3.1).	Each source plays at its rate with no dropped vsyncs; the frame-ID strip decodes from a photograph; ego speed matches the clip (a known-speed stretch).
T-D	Perception	`laned`, `detectd`, `planed`, `scored`; the pitch calibration per recording (8.4); the lane-network choice by score. Produces measured per-job times (Table 5.2 filled in).	§5.7 on 300 MetaDrive frames; times measured; agreement with `modelV2` reported.
T-E	Registration, sync modes	The mirror homography (8.3); `hudd` in truth-fed sync mode, then perception-fed sync mode. Produces the display chain's own residual and the latency-free residual of the whole chain.	Homography ≤ 2 px RMS; truth-fed: drawn line within 0.10° of the painted line's truth at 10, 20, 30 m, median over the 24 s MetaDrive drive, chevron likewise against the lead; perception-fed: residual reported against the §5.7 angles.
T-F	Latency, live mode	The Pico probe with two photodiodes; monitor lags; the chain; the +200 ms sign test on a comma2k19 curve; compensation on and off against MetaDrive truth at t_{photon}.	Chain ≤ 90 ms median (p95 recorded); sign confirmed both ways; compensated lead residual (p95) below uncompensated. Closes B6's G2 except "runs on the Jetson".
T-G	Photometry	Metered whites and blacks, the glass's measured split, the ghost in both orientations, the gains recomputed (8.5); the ribbon fill share, line widths and night level judged at the plan's ratios.	Measured ratio within ±10% of each set condition; ghost ≤ 2% or documented; glass orientation recorded.
T-H	Parallax variant	The slide to Δ = 31.6 mm; the webcam tracker (8.7); the rail test at ±75 mm. Produces the plan's head-motion number without a car.	≤ 0.3° with the tracker on; the same run with it off shows ≈ 0.36°.

STAGE	BUILDS	NEEDS · PRODUCES	GATE
T-I	Camera in the loop	A webcam or the OAK-D Pro, later the ZED on the Jetson; the loop homography (8.8); `roadd` 's capture with the plane fit off. Produces the camera's own latency and its cost in lane residual.	Homography ≤ 2 px; residual within 0.05° of the direct path; latency through the camera measured. When the Jetson runs it: G2 complete.

11.1 Acceptance

STAGE	PASSES WHEN	MEASURED BY
T-A rig	Test card reads correctly; parallax check: cross on the PGU and cross on the road monitor separate by ≤ 5 px over a 100 mm eye shift (planes within 2.8 mm)	eye camera, two photographs
T-C displays	Both monitors at 3,840 × 2,400, 60 Hz, native pixels, no scaling, no auto-brightness; each full-screen window on its own monitor	screenshot and the monitor's own menu
T-E calibration	Homography residual ≤ 2 px RMS (0.017°) over the whole window; flip correct	eye camera, one photograph
T-D perception	Plan §5.7 on 300 MetaDrive frames with truth: ego lane found ≥ 95%; lateral error ≤ 0.15 m at 10 m, 0.30 at 20, 0.50 at 30; vehicles within 50 m ≥ 90%; pedestrians within 30 m ≥ 85%	scored against truth
T-E sync modes	Truth-fed: drawn lane line within 0.10° (11 px) of the painted line's truth on the road monitor at 10, 20 and 30 m, median over a 24 s MetaDrive drive. Perception-fed: the same residual reported, expected within the §5.7 angles (0.86°, 0.86°, 0.95°)	hudd's own residual against truth
T-F live mode	hudd at 60 fps with no dropped frames for 10 min; road photon to HUD photon ≤ 90 ms (median and 95th); prediction sign confirmed with the +200 ms test; lead chevron 95th-percentile residual with compensation below the value without	photodiode pair; residual log
T-G photometry	Measured contrast at the eye within ±10% of the set condition for ratios 1.15, 1.19 and 1.39; ghost ≤ 2% of the main image or documented	spot meter or calibrated camera
T-H parallax variant	With Δ = 31.6 mm and ±75 mm sway: residual ≤ 0.3° with the tracker on; the same test with it off fails (≈ 0.36°)	eye camera on a rail
T-I camera loop	Camera-to-monitor homography ≤ 2 px; lane residual through the camera within 0.05° of the direct path; latency through the camera measured	as above, with the loop camera

Table 11.1. What passes each stage and how it is measured. Every number traces to a plan criterion or to this document's derivation; none is waived, a failed gate is recorded with its number.

11.2 What the bench does not verify, so that no one thinks it did

- Any optical property of Stage 1 or Stage 2: image distance, field, eye box, mirror figure, astigmatism, the windshield's ghost, polarisation at 65°, the solar focus.
- Luminance at the eye in daylight, as an absolute: the bench reproduces ratios up to 1.43 against a background of 840 cd/m², not 10,000.
- Stereo range and the per-frame plane fit; the eye-side range error of the plan's §5.4 (the camera-to-eye baseline is zero on the bench).
- GNSS latency, the F9R, the IMU fusion, dead reckoning.

- The controller, dimming, the shutter, the backlight, heat, vibration, the installation checklist.
- The lane network's generalisation to the demo course: comma2k19 is one highway in California; the phone clips of T-C should be of the roads the demo will use.

11.3 Order and time

T-A and T-B are a few days each and independent; T-C needs the comma2k19 chunk downloaded (10 GB) and the MetaDrive exporter written; T-D is the long one, a week or two of network exports, scoring and the pitch tool; T-E and T-F are a day each once T-D holds; T-G needs the meter; T-H the webcam; T-I the camera and, for its full value, the Jetson. The plan's week-1 measurements (GPU time per job) come out of T-D on the Mac and are repeated on the Jetson under B2.

12 Risks, gotchas and the bill of materials

12.1 Things that will go wrong, and what to do

SYMPTOM OR RISK	CAUSE AND RESPONSE
The overlay slides against the road when the viewer moves	Planes not coplanar (section 3.1). Shim the road monitor or the PGU until the parallax check passes; a rail stop that drifts is the usual culprit.
The reflection is dim, far under 30% of the PGU's white	The PGU's polarisation is p to the glass (section 3.5). Retarder film, or rotate the PGU 90° and the picture in software; meter again.
Every line is doubled, 16 px apart	The glass's bare face is toward the viewer. Turn the glass over; if both faces are bare, accept the ghost on the bench and use thick lines, as the plan's windshield rule does.
The road picture is blurry at the eye	Expected for comma2k19 (× 7.2); use the comma three route or a phone clip for demos; perception is unaffected.
The picture is at half resolution or offset on a monitor	macOS scaling: the window was created at logical, not physical, pixels, or the display is set to "larger text". Use the high-DPI flag and native resolution; verify by photographing a 1 px checkerboard.
Judder on the road monitor	A 20 Hz source on a 60 Hz display shows each frame for three vsyncs; a dropped vsync shows as a 4-vsync frame. `display` logs every schedule; look for a slow decode (software HEVC path) and force the hardware decoder.
hudd drops frames	Vsync stacking or a compositor frame; the warp mesh rebuilt every frame; another window on the PGU's display. B6.8 applies.
Chain over 90 ms	Find the stage with the probe's split measurement (byte-to-patch for render plus PGU lag; vsync-to-patch for the road monitor's lag). A portable monitor with two frames of scaler lag at each end eats 67 ms by itself; if so, record it and note that the car's HDMI board is a different device.
Compensation makes it worse	The sign (B6.5), or `hudd`'s target set to t_{photon} while measuring against the picture (section 6.2): switch the target to the picture's time for registration tests.
Lane residual grows with range on comma2k19	Pitch or height wrong (Table 5.1). Refit the vanishing point on a straight stretch; try the height as a free parameter against the lane width (3.7 m) at 20 m.
The networks score well on MetaDrive and badly on real clips, or the reverse	Expected; MetaDrive's paint is synthetic and its score is a floor. Weight the real-clip agreement with `modelV2` and the hand-checked frames; the plan's fine-tuning decision belongs to G1.
The PGU gets hot face up in the well	1,200 nits in a closed well with a hood. Leave the well's ends open, run the PGU at the gain the condition needs rather than at full white, watch for dimming.
A monitor drops out on the Mac's port	Power: use the monitors' own PD chargers; check which ports of the MacBook support two displays.
The Mac's inference times are nowhere near the Jetson's	They are not meant to be; the bench times size the bench's own chain. The plan's budget is measured under B2 on the Jetson.
The loop camera sees the HUD	It is inside the reflection's cone: raise it above the glass's top edge, or hood the gap between glass and camera.
Two hosts disagree on time	chrony not synced, or a topic stamped with the wrong clock. The photodiode chain is clock-free; use it to find which host's stamps are off.

12.2 Bill of materials

ITEM	LOW (USD)	HIGH (USD)	BASIS
YODOIT 16-inch 3840 x 2400 monitor, unit 1 (PGU, face up)	0	0	owned
YODOIT 16-inch 3840 x 2400 monitor, unit 2 (road, upright)	0	0	owned
MacBook Pro M4 (drives both monitors, runs everything until the Jetson is back)	0	0	owned
Neewer NA17 beamsplitter glass, 315 x 315 x 2 mm, 70T/30R, AR coat	53	53	neewer.com; the plan's cheap teleprompter glass
Enclosure: 6 mm plywood or black foam board, flock paper, glue	25	45	estimate
Slide for the road monitor: 300 mm drawer slide or macro rail, with a mm scale	15	45	range
Chin rest or camera tripod with a quick-release at the eye point	0	40	owned tripod or a cheap chin rest
Latency probe: Pico, two BPW34 photodiodes, LM393, resistors	10	25	B6.1 parts, one photodiode more
Chessboard and ArUco prints on card, 1 mm ruler, angle gauge	5	15	estimate
Webcam, 1080p 30 fps, for the head tracker on the bench	25	80	or the OAK-D Pro when it arrives
Half-wave retarder film sheet, only if the polarisation test calls for it	0	150	plan BOM; optional
USB-C PD charger for each monitor (the Mac's ports may not hold 1,200 nits)	0	40	verify what the monitors draw
Total of new purchases	**133**	**493**	

Table 12.1. New purchases for the bench. The plan's spot meter, the OAK-D Pro and the ZED are not repeated here; the bench uses them when they arrive and works without them until then.

12.3 Decisions this design leaves open

- Eye-to-image distance: 600 mm by design; Table 3.1 for the others. Settled at T-C by looking.
- Glass orientation (coated face): settled by the ghost photograph at T-A or T-G.
- Whether the PGU needs the retarder film: settled by the photometry at T-G.
- Which lane network: settled by score at T-D, as the plan intends.
- Whether a larger glass is worth buying so the whole PGU reflects: only if a wider window than the plan's is wanted on the bench.
- When the Jetson joins: after B2 (reflash); T-I is written so that it can run on the Mac with a webcam first.

A Formulas

QUANTITY	FORMULA AND THE VALUES USED
Image of a PGU point	Mirror about the plane $z = -x$ through the glass centre: (x, y, z) maps to (–z, y, –x). The PGU face $z = -200$ maps to $x = 200$, the road monitor's face.
Life-size field	$2 \cdot \operatorname{atan}(W / 2D)$ with W the monitor's active width and D the eye-to-image distance: 32.7° at 600 mm.
Window on the monitors	$2 \cdot D \cdot \tan(\text{fov} / 2)$: 113.9 × 56.9 mm; divided by the pitch, 1,242 × 621 px; 115 px per degree.
Life-size crop of a source	$2 \cdot f \cdot \tan(\text{field} / 2)$ source pixels across the shown field; the scale to the monitor is 3,840 divided by that.
Coplanarity by parallax	Separation on the far plane $\approx e \cdot \delta / D$ for an eye shift e and a plane offset δ: 1.8 px per mm at e = 100 mm.
Parallax variant	Angle per sway e: $e \cdot (1/D - 1/(D + \Delta))$. For the car's parallax between 7.5 m and z: $k = 1/7.5 - 1/z$; $\Delta = 1/(1/D - k) - D$.
Contrast ratio and gain	ratio = $(L_{HUD} + L_{bg}) / L_{bg}$; gain = $(\text{ratio} - 1) \cdot L_{\text{bench bg}} / L_{\text{PGU white at the eye}}$.
Fresnel at 45°, n = 1.52	$\theta_t = 27.7°$; $R_s = 9.7\%$, $R_p = 0.94\%$.
Ghost separation	$2 \cdot t \cdot \tan \theta_t \cdot \cos \theta_i = 1.49$ mm for t = 2.0 mm; 8.5′ at 600 mm; relative brightness $T^2 \cdot R_2 / R_1$.
Lifting	$\alpha = \operatorname{atan}((v - c_y) / f)$; $z = h / \tan(\theta + \alpha)$; $y = z \cdot (u - c_x) / f$.
Age compensation	$p_{now} = R_z(-\psi) \cdot (p_{capture} - [s, 0, 0])$, s the integral of v and ψ the integral of ω over the age; tracks also by their own velocity.
Bench projection chain	PGU pixel = $H_{PR} \cdot F \cdot S \cdot K_{source} \cdot \text{camera_from_car} \cdot p_{now}$.

B The calibration file

```
rig:
  frame: "x along the viewing axis (eye negative), y right, z up, origin at the glass centre, mm"
  d_glass_mm: 200          # glass centre to the PGU face and to the road monitor face (measured)
  d_eye_mm: 400             # eye to the glass centre (measured)
  delta_mm: 0                 # road monitor slid back (parallax variant)
  glass: {w_mm: 315, t_mm: 2.0, tilt_deg: 45, coated_face: "pgu"}      # orientation by the ghost test
monitors:
  pgu:  {name: "YODOIT unit 1", edid: "", px: [3,840, 2,400], pitch_mm: 0.0917, white_cdm2: 1200, lag_ms: null}
  road: {name: "YODOIT unit 2", edid: "", px: [3,840, 2,400], pitch_mm: 0.0917, white_cdm2: 1200, lag_ms: null}
warp:
  flip: "vertical"
  homography_pr: [[-1, 0, 0], [0, 1, 0], [0, 0, 1]]     # road-monitor px -> PGU px, refined by tools/calib_rig.py
  residual_px: {rms: null, max: null, date: null}
window:
  fov_deg: [10.8, 5.4]
  px: [1,242, 621]
  lookdown_deg: 5
sources:
  comma2k19:
    camera: {f_px: 910, cx: 582, cy: 437, size: [1164, 874], distortion: null}
    mount:  {height_m: 1.22, pitch_deg: null, yaw_deg: null, lateral_m: 0}     # pitch per segment from the vanishing point
    crop:   {src_px: [534, null], scale: 7.2}
  oproute:
    camera: {f_px: 2,648, cx: 964, cy: 604, size: [1928, 1208]}
    mount:  {from: "liveCalibration"}
  metadrive:
    camera: {f_px: 6,543, size: [3,840, 2,400]}
    mount:  {from: "export"}
photometry:
  glass: {t_road: null, r_pgu: null, ghost_rel: null, ghost_px: null}
  gains: {plan_min_sky: 0.35, bl_sky: 0.45, bl_road: 0.18, s1_sky: 0.084}  # recomputed after metering
timing:
  lag_road_ms: null
  lag_pgu_ms: null
  render_ms: null
  chain_ms: {median: null, p95: null, date: null}
tracker:
  webcam: {f_px: null, cx: null, cy: null, pose_rig: null}
  residual_mm: null
loop:
  camera: {f_px: null, cx: null, cy: null}
  homography_cam_road: null
  residual_px: null
```

C Sources and provenance

- The plan: `docs/ARHUD-PLAN-TFT.pdf` and `docs/ARHUD-PLAN-TFT-SPEC.md`; numbers from `docs/arhud-plan-tft-src/calc.py` (window 10.8° × 5.4°, monitor 277 ppi and 1,200 cd/m², the 70/30 combiner, the latency budget, H_CAM = 1.22 m, the ZED's focal length).
- The build guide, stage B6: `docs/ARHUD-BUILD-GUIDE.pdf` (the Stage 0 rig, hudd's two passes, the probe, the sign test, the drawing rules).
- The road simulator: `docs/arhud-road` (truth, the registration bench, the linear-light compositing, the 39% ribbon finding, the head-sway results).
- Camera models: `openpilot/common/transformations/camera.py` in this checkout (`_neo_config` 1164 × 874 f = 910; `_ar_ox_config` 1928 × 1208 f = 2648); frame and log readers in `openpilot/tools/lib`; the demo route in `openpilot/tools/clip/run.py`.
- comma2k19: github.com/commaai/comma2k19 (MIT; segment layout; processed_log fields; ~100 GB in ~10 GB chunks on Academic Torrents), read 2026-10-27.
- Neewer NA17 beamsplitter: neewer.com product page (31.5 × 31.5 × 0.2 cm, 70T/30R, tempered, anti-reflection coating, $52.99), read 2026-10-27.
- YODOIT 16-inch 3840 × 2400 monitor: vendor listings (1,200 nits; contrast 1,200:1 or 2,000:1 depending on the listing; USB-C), read 2026-10-27; HDMI input inferred from the Jetson having driven one on 2026-10-07.
- Fresnel reflectances from the plan's formula at n = 1.52; ghost geometry from the plan's §7.3 applied to 2 mm glass at 45°.
- Hardware state on 2026-10-27: MacBook Pro (Mac16,1, Apple M4, 24 GB), one YODOIT attached at 3840 × 2400 60 Hz; the Jetson unreachable by its Tailscale name and by mDNS; no CUDA on the mini PCs; venvs `arhud-cad` (3.12) and `arhud-metadrive` (3.11) present, neither with OpenCV, PyTorch or ONNX Runtime.